

**Bi Objective Multi Depot Location Routing Problem with
Time Window for EVs**

by

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The above committee determined that the thesis is acceptable in form and content and that a satisfactory knowledge of the field covered by the thesis was demonstrated by the candidate during an oral examination. A signed copy of the Certificate of Approval is available from the School of Graduate and Postdoctoral Studies.

Abstract

This thesis addresses a multi-depot, bi-objective Location Routing Problem for electric vehicles (EVs), modeled as a Mixed Integer Linear Programming (MILP) problem. The mathematical model integrates cost minimization with workload fairness under time windows. A custom metaheuristic algorithm, combining a hybrid Genetic Algorithm (GA) and Ant Colony Optimization (ACO), was developed to optimize depot locations and EV routes, minimizing operational costs and balancing driver workloads. The approach accounts for depot and vehicle capacities, vehicle mileage limits, and EV charging constraints. Computational experiments reveal that the model's efficiency outperforms traditional solvers for large-scale problems, maintaining solution quality within a 5% gap. A Pareto analysis provides insight into cost vs. distance trade-offs, and sensitivity analysis examines the impacts of vehicle capacity and route distance constraints. This work contributes to sustainable logistics by offering a practical, scalable solution for EV routing, enhancing real-world deployment feasibility in urban settings.

Keywords: Electric vehicle; Multi depot location routing problem; Workload Fairness; MDLRPTW

Author's Declaration

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Chapter 1

Introduction

The logistics and transportation industries have undergone major changes due to the rapid expansion of e-commerce and growing environmental concerns. To address these challenges, many companies are turning to electric vehicles (EVs) for their delivery services, aiming to lower carbon emissions, cut operational costs, and reduce reliance on fossil fuels (Anosike et al., 2023; Prohaska et al., 2016; Zhao et al., 2016). EVs provide several benefits, including reduced greenhouse gas emissions, decreased noise pollution, and access to government incentives, making them an appealing choice for urban delivery services (Pelletier et al., 2016).

As urban populations grow, the demand for transportation services rises, intensifying environmental issues. Emissions from traffic, including noise and air pollutants, play a major role in worsening urban air quality. This is particularly concerning since nearly one-third of urban traffic is attributed to commercial activities (Hebes et al., 2010). The environmental challenges caused by increased traffic are worsened by the growing need for goods transportation, mainly driven by the rapid growth of online shopping (Ehrler et al., 2021). The rise of online shopping has significantly boosted the demand for deliveries and returns, a trend that continues to grow (Morganti et al., 2014). As more consumers turn to online services, such as grocery shopping, the logistics industry faces

the challenge of optimizing delivery networks to handle increasing demand while reducing environmental impact (Hanus, 2016).

As a response to these challenges, one of the most widely studied optimization models is the Vehicle Routing Problem (VRP), which seeks to find the most efficient routes for vehicle fleets to serve geographically dispersed customers at the lowest cost and environmental impact possible. The VRP is a classic logistics challenge involving determining the most efficient routes for a fleet of vehicles to deliver goods to a dispersed customer base. Originally introduced by Dantzig and Ramser as the "Truck Dispatching Problem" (Dantzig and Ramser, 1959), the VRP aims to minimize the costs and distances associated with deliveries. Over time, the problem has evolved to address complex modern challenges, such as urban congestion, time windows, and sustainability (Toth, 2002; Toth and Vigo, 2014).

Incorporating EVs into logistics systems introduces specific challenges, particularly in routing and facility location decisions, due to EV-related limitations such as restricted driving range, lengthy charging times, and limited charging infrastructure. Schneider et al. (2014) addressed these challenges by developing the Electric Vehicle Routing Problem with Time Windows and Recharging Stations (E-VRPTW), which integrates the need for EVs to stop at recharging stations to overcome range constraints. Charging times for EVs can vary significantly, according to the US Department of Transportation, ranging from approximately 45 minutes at ultra-fast charging stations to over 10 hours at slower chargers. These variations can lead to delays and impact customer satisfaction and time windows. Schneider et al. (2014) highlight how these charging durations affect logistics operations, emphasizing the importance of careful planning to minimize disruptions. In addition, the scarcity of public charging stations can result in long queues, further complicating scheduling and delivery operations. Davis and Figliozzi (2013) examine the effects of limited charging infrastructure on freight logistics, underlining the need for

well-planned charging networks to facilitate the broader adoption of EVs in logistics. Electric delivery vehicles generally have a more limited driving range compared to traditional gasoline-powered vehicles before requiring a recharge. EVs are mainly classified into two types: Battery Electric Vehicles (BEVs) and Hybrid Electric Vehicles (HEVs). BEVs can cover distances ranging from 150 km to 450 km on a single charge, depending on factors like the type of vehicle, its load, and environmental conditions. This makes driving range a crucial consideration in route optimization (Feng and Figliozzi, 2013; Schneider et al., 2014). Recent advancements in EV technology have introduced several advantages over earlier models, including multiple charging methods. For example, ultra-low overnight charging takes approximately 8 to 10 hours to achieve a full charge, benefiting from lower electricity rates and reduced costs. In contrast, fast chargers can charge a battery up to 80% in about 40 minutes, although they are less common and mainly found in developed cities (Hydro One, 2024).

In Electric Vehicle Routing Problems (EVRPs), a common approach is to minimize the total number of EVs used, followed by minimizing the total distance traveled (Schneider et al., 2014). This approach can result in significant cost savings by reducing fixed expenses and labor costs associated with BEVs. However, relying on fewer vehicles might lead to longer travel distances for each vehicle, which could affect efficiency and strain battery limitations. Optimizing their routes is essential to ensure they can cover the required distances more effectively (Schneider et al., 2014; Schiffer and Walther, 2017). BEVs are particularly well-suited for shorter distances and urban environments. Their efficiency at lower speeds, reduced noise levels, and ability to handle frequent stop-and-go traffic make them an ideal choice for city deliveries. Additionally, financial incentives for EVs add to their appeal. For instance, BEVs are practical for delivery routes averaging shorter distances, such as the 68 kilometers typically traveled by FedEx drivers in the United States (Feng & Figliozzi, 2013). In such cases, BEVs can handle deliveries and recharge once they return to the depot (Van Duin et al., 2013).

One of the key challenges in EVs routing lies in determining the optimal locations to implement charging stations or depots. These locations must balance operational efficiency, minimize travel and charging costs, and meet constraints such as vehicle range limitations and customer time windows, all while ensuring adequate coverage for the fleet's recharging needs.

Webb (1968) and Christofides and Eilon (1969) were among the first to identify inaccuracies in location modeling caused by the out-and-back approach to calculating delivery costs. By the mid-1970s, studies on location-routing problems (LRP), including models, solution methods, and practical applications, started to emerge in academic literature. Laporte (1988) provides an overview of the research on LRP that was published before 1988. Worley et al. (2012) introduced one of the earliest models for the LRP specifically for EVs. Their model aimed to minimize the total costs of travel, recharging, and establishing recharging stations, though they did not provide a solution method. Later, Yang and Sun (2015) tackled similar challenges by integrating facility location problem models (FLP) with VRP. Paz et al. (2018) expanded existing LRP models for EVs by introducing a version that incorporates multiple depots. Their model also includes critical features such as time window constraints, making it more applicable to complex logistics systems. The multi-depot routing problem (MDVRP) is an important extension of the traditional VRP, where customers are served from multiple depots (de Oliveira et al., 2016). The goal of the MDVRP is to minimize the total distance or time required to serve all customers, as shorter delivery times lead to higher customer satisfaction (Montoya-Torres et al., 2018). The MDVRP (Pardalos & Coleman, 2009) extends the Single-Depot Vehicle Routing Problem (SDVRP) by allowing vehicles to operate from multiple depots, with each vehicle returning to its starting depot at the end of its route (Montoya-Torres et al., 2018).

In real-world applications, customer demands frequently change due to factors like location and seasonal variations. The MDVRP seeks to optimize delivery by minimizing the

total distance traveled and service time required to meet these demands (Jayarathna et al., 2020). Additionally, it can help reduce the number of vehicles needed, along with associated labor and operating costs (Fitriana et al., 2019). Li, Li, and Pardalos (2016) introduced an MDVRPTW, which integrates multiple depots into the VRPTW framework. In this model, vehicles are supplied by different depots, and their tasks must begin within specified time intervals.

This problem is modeled as an integer nonlinear multi-commodity network flow and is solved using a column generation technique integrated within a branch-and-bound framework, forming a branch-and-price approach. Building on this, Bae and Moon (2016) applied the MDVRPTW to a real-world logistics and supply chain management challenge, specifically focusing on using vehicles for the delivery and installation of electronic equipment.

This paper focuses on tackling key challenges in optimizing the LRP in urban areas to achieve efficient and sustainable delivery operations. We propose a model that determines the best depot locations with charging stations, ensures timely delivery to customers within specified time windows, and minimizes overall costs associated with depots and EVs. These costs include travel distance, depot installation cost, vehicle expenses, and considerations for workload fairness. Workload fairness is a critical consideration in optimizing vehicle routing, as it ensures that all drivers are treated equally and provided with equal opportunities to perform their duties and earn a living wage, which is essential for fostering equity in the transportation industry (Jütte et al., 2017; Ma et al., 2017). There is no doubt that implementing fair scheduling practices will enhance driver satisfaction and reduce turnover rates, contributing to a more sustainable and efficient transportation system (Hamdan et al., 2023). Besides improving operational efficiency, workload fairness has a broader impact. Equitable workload distribution can prevent strikes and financial losses while ensuring customer satisfaction by avoiding delivery delays. For instance, in April 2024, Amazon delivery drivers and dispatchers staged the

first-ever strike by Amazon drivers in the U.S., protesting unfair labor practices and unsafe working conditions. This action underscored how unfair workloads disrupt services, leading to delayed deliveries and dissatisfied customers.

By incorporating workload fairness, this study aims to address these challenges, providing a framework that balances operational efficiency with equity for drivers. Gansterer and Savelsbergh (2018) expanded on the collaborative pickup and delivery problem introduced by Berger and Bierwirth (2010), introducing workload constraints for scenarios involving a single vehicle and a single time period. Gansterer et al. (2021) further advanced research on collaborative transportation systems by addressing multi-vehicle scenarios in multi-depot settings, incorporating workload constraints to distribute tasks more evenly across drivers. While their study did not consider time window constraints, it highlighted the importance of equitable workload distribution in improving efficiency and fairness. Their methodology relied on the Adaptive Large Neighborhood Search (ALNS) implemented in C++. Following this, Soriano et al. (2023) extended the focus to profit fairness in the MDVRP, ensuring coalition partners gained profits proportional to their standalone operations. Like Gansterer et al., their study utilized ALNS but did not address real-world scenarios, such as EV consumption, customer time windows, or operational costs.

Building on these studies, our research integrates workload fairness into the MDLRPTW, aiming to minimize total costs while maintaining fairness among drivers. Ensuring equitable workload distribution not only optimizes vehicle routes between depots and customers but also prevents EVs from depleting their batteries during operations, adding another layer of efficiency to the model. Our study is important because we model a practical MILP framework and develop a hybrid metaheuristic algorithm while explicitly addressing computational time for different problem sizes. Unlike previous studies, which do not analyze or report the computational time required to solve large-scale problems, our research focuses on the scalability and computational efficiency of the proposed

approaches. Our study is therefore relevant to real-world applications, where solving large-scale problems efficiently is important.

In this study achieves an optimal solution that balances minimizing total costs with reducing the maximum travel distance for each vehicle, ensuring fairness among drivers. The model incorporates travel time, energy consumption, and customer time windows to provide a practical approach to logistics optimization. Using the General Algebraic Modeling System (GAMS) with the CPLEX solver, we solve the problem as a MILP model for small-scale instances to obtain exact solutions. These solutions are then compared with those obtained from a hybrid metaheuristic approach, which combines a Genetic Algorithm (GA) with Ant Colony Optimization (ACO) and is implemented in Python for large-scale problems.

Chapter 2

Literature Review

2.1 Adoption and Challenges of Electric Vehicles in Logistics

The adoption of EVs in logistics is driven by their potential to mitigate environmental impacts associated with conventional internal combustion engine vehicles. Studies have demonstrated that EVs can significantly reduce carbon dioxide emissions and other pollutants, aligning with global initiatives to combat climate change (Hawkins et al., 2013). As well as being environmentally friendly, EVs also offer significant economic and operational benefits. EVs, which have lower energy costs because they use fewer moving parts than combustion engines, and lower maintenance expenses, which increase their competitiveness (Feng and Figliozzi, 2013). As EVs emit no tailpipe emissions and generate no noise, they are ideal for urban logistics since they are well-suited to densely populated areas (Verma et al., 2022). Moreover, many countries offer incentives to encourage the adoption of EVs, including tax reductions, subsidies, and financial incentives (Mpoi et al., 2023). Recent advancements in battery technology and economies of scale have further reduced costs, making EVs more accessible for logistics operations (Stamadianos et al., 2023).

Despite these benefits, integrating EVs into logistics operations presents several challenges. Limited driving ranges, shorter than those of diesel vehicles, necessitate meticulous route planning to prevent battery depletion (Pelletier et al., 2016). The uneven distribution and scarcity of charging stations exacerbate operational difficulties, particularly in regions with inadequate infrastructure (Davis & Figliozzi, 2013). An EV's payload directly impacts its range. Since payloads have more inertia than empty vehicles, vehicles with a payload will require more energy to move. Typically, EVs have a lower payload rating than ICE vans (Kyriakakis et al. 2022). Additionally, longer charging times compared to conventional refueling can complicate scheduling and lead to delays. Partial recharging, when compared with full recharging, has no considerable difference in terms of total operational costs (Schneider et al., 2014; Zang et al. 2022).

The incorporation of EVs into logistics operations impacts routing decisions significantly. Traditional Vehicle Routing Problems (VRPs) must be extended to account for energy consumption, battery capacities, and charging requirements. Ignoring these factors can lead to infeasible routes or increased operational costs due to unplanned detours or delays. The literature emphasizes the importance of strategic planning in EV infrastructure, which involves balancing the costs of installation and operation with the benefits of widespread coverage and accessibility (Gnann et al., 2018).

2.2 Evolution of Location Routing Problems (LRP) in Modern Logistics

Facility location and vehicle routing are among the most critical problems in logistics network design. In the past, these two combinatorial optimization problems have been solved separately by determining the location of facilities, and then planning the routes after. VRP is first introduced by Dantzig and Ramser as the "Truck Dispatching Problem" (Dantzig and Ramser, 1959), The VRP with Time Windows (VRPTW) introduces

scheduling constraints by requiring deliveries to occur within specific time frames, adding complexity to route planning and vehicle dispatch (Desrochers et al., 1992). This variant is crucial in industries where timely deliveries directly affect customer satisfaction and operational efficiency Kirci (2016) introduces a CVRPTW model utilizing Tabu Search and Hopfield Neural Networks to optimize routing and minimize costs, demonstrating improved performance through real-world scenarios and comparisons with Simulated Annealing. Bae and Moon (2016) extend the Multi-Depot Vehicle Routing Problem with Time Windows (MDVRPTW) by incorporating delivery and installation vehicles, addressing coordination between product delivery and installation services. The model minimizes depot, vehicle, and labor costs while satisfying dual time constraints. A heuristic and genetic algorithm are developed to find near-optimal solutions, demonstrating efficiency in solving large-scale problems to reduce vehicle emissions and dependence on fossil fuels, Electric vehicles (EVs) have become the leading lower-carbon transport equipment (Riemann et al., 2015).

Electric Vehicle Routing Problems present unique challenges, primarily due to the limited range and battery capacity of EVs. This necessitates careful route planning to ensure vehicles either operate within their range or make scheduled charging stops (Schneider et al., 2014). Effective charging strategies play a major role in this process, requiring decisions on when and where to charge while factoring in charging times and station availability (Conrad & Figliozzi, 2011). Shao et al. (2018) developed an EVRP that minimizes vehicle, travel, and charging costs by assuming constant charging times and a full recharging policy. A hybrid genetic algorithm was developed to address the problem in which genetic algorithms are combined with local search techniques in order to solve this issue. To address these challenges, researchers have developed a variety of models, including time-dependent EVRPs that account for fluctuating travel times and energy consumption rates (Hiermann et al., 2016). The EVRPTW was extended by Desaulniers et al. (2016) to introduce flexible recharging options and one charging stop per route.

Further advancements in EV charging technologies have aimed to enhance efficiency and minimize downtime. These include regular, fast, and ultra-fast charging systems (Felipe et al., 2014; Keskin & Çatay, 2018). Additionally, partial charging strategies have been introduced to improve operational flexibility and reduce charging durations (Keskin & Çatay, 2016). A rising interest in MDEVRPTW has emerged in recent studies, driven by increasing attention to EV technology and its adoption in business operations (Kucukoglu et al., 2021).

Based on the study by Von Boventer (1961), the idea of combining the location and routing problems was first conceived. These traditional approaches lead to a suboptimal result, given the fact that they are heavily linked to each other and with the passage of time, this evolved into the LRP as well (Laporte, 1988; Salhi and Rand, 1989; Salhi and Nagy, 1999). The LRP is a strategic decision-making problem that combines facility location and vehicle routing decisions to minimize overall logistics costs which addresses two critical questions: where to locate depots or facilities and how to route vehicles to serve customers efficiently from these locations (Nagy and Salhi 2007).

Classical LRP studies typically focus on cost minimization, encompassing facility opening costs, transportation costs, and vehicle operating costs (Prodhon & Prins, 2014). However, subsequent efforts have extended the traditional LRP by incorporating more realistic elements such as heterogeneous fleets, capacity constraints, and multi-echelon structures (Eskandarpour et al., 2015). Drexler and Schneider (2015) identify three situations where specific conditions apply to facility selection. These include: (1) when opening a facility incurs either fixed or variable costs, (2) when there is a requirement to select a certain number of facilities, or a maximum limit exists on how many can be chosen from a larger set, and (3) when facilities have capacity constraints. In order to address the difficulties of LRP research, real-world constraints are increasingly included, such as time windows constraints. Time window constraints, which define the permissible time frames for deliveries or pickups based on customer availability or service agreements,

introduce significant complexity to LRPs(Desaulniers et al., 2002; Ponboon et al., 2016). Worley and Klabjan (2012) introduced one of the first models for the LRP of EVs, aiming to minimize travel, recharging, and station location costs. However, they did not propose a solution method. Some studies aims to determine the location of battery swap stations, as well as developing a routing plan for EVs, to assist in determining the locations of the stations(Yang & Sun 2015; Hof et al., 2017). Schiffer and Walther (2017) introduced an LRP model that considers conventional charging stations and allows for partial battery charging. The model is efficient for small instances but struggles with larger ones. To address this, they later developed an Adaptive Large Neighborhood Search algorithm for large-scale problems to present a robust LRP using EVs with time windows and partial recharging. There is uncertainty in customers' spatial distribution, demand, and service times(Schiffer and Walther, 2018). The algorithm was validated using instances from Yang and Sun (2015) and Schiffer and Walther (2017).

Hiermann et al. (2016) introduces an analysis of a charging station LRP model with mixed fleet sizes in this study, in which the competitiveness of electric and traditional vehicles is compared and the problem is solved using branch-and-price methods. Multi-objective combinatorial optimization of charging facilities is a complex combinatorial optimization problem. There are several studies has delved into multi-objective LRP models for EV charging infrastructure, addressing various key objectives such as minimizing costs (Wang et al., 2020), maximizing profits (Yücel et al., 2020), enhancing coverage (Bao & Xie, 2021), improving customer satisfaction (Alamatsaz, 2022), and reducing environmental impacts (Tavana et al., 2021). For example, Zu and Sun (2022) focused on optimizing charging station placement by analyzing user behavior to both elevate user satisfaction and lower energy consumption during vehicle distribution.

Multi-depot LRPs involve determining the optimal number and locations of depots and assigning customers to these depots while planning vehicle routes. The problem's complexity increases with multiple depots due to the expanded combination of possible depot-

customer assignments and routing options (Cordeau et al., 1997). Facility location decisions are integrated into the MDVRP with Pickup and Delivery (MDVRPPD), forming the MDLRP with Pickup and Delivery (MDLRPPD). This approach aims to optimize both vehicle routing and facility location simultaneously (Capelle et al., 2019). Paz et al. (2018) present three EV MDLRP models, considering battery swapping (BSS) and conventional charging. The MDLRP models optimize small-scale problems and compare charging strategies to refine network design. Results show installation and operational costs significantly affect decisions, highlighting the need to consider station technology and real-world data in future large-scale applications.

Portela et al., (2021) presents two constructive methods, the Cheapest Insertion and Disruption of Routes, to solve the Multi-Depot Electric Vehicle Location Routing Problem with Time Windows and Battery Swapping (MDEVLRPTW-BS). These methods are combined with GRASP and RVND algorithms to optimize EV routing by selecting depots, assigning vehicles, and locating battery swap stations, resulting in improved performance and reduced processing times for large-scale logistical problems. Du et al. (2023) propose a Multi-Depot Two-Echelon Joint Delivery LRP (MD-2E-JDLRP) to reduce carbon emissions by integrating EVs into shared delivery networks. Their hybrid heuristic algorithm enhances EV routing and depot placement, boosting sustainability in last-mile logistics.

2.3 Workload Fairness in Logistics

Workload fairness is a critical component in logistics operations, influencing driver well-being, operational efficiency, and the long-term sustainability of supply chains. Ensuring equitable workload distribution helps mitigate driver fatigue and stress, reducing burnout and turnover, which are key factors in maintaining stable and productive fleets (Xiang et al., 2008).

Studies by Mancini et al. (2021) and Hezarkhani et al. (2016) highlight the significance of workload balancing in collaborative vehicle routing and fair cost allocation, reinforcing the notion that fairness is essential for fostering long-term logistics partnerships. Equitable distribution not only upholds ethical labor standards but also enhances resource utilization by preventing vehicle idleness and ensuring efficient use of depots. Additionally, compliance with labor regulations regarding driver hours promotes safety and aligns operations with legal frameworks, minimizing the risk of penalties and legal disputes (Carter & Rogers, 2008). Strategies to achieve fairness often involve minimizing the maximum distance traveled by any vehicle (Koç & Bektaş, 2014) or balancing route durations (Lim & Zhang, 2007). These approaches are commonly framed as multi-objective optimization problems that seek to reduce costs while equalizing workloads (Jozefowicz et al., 2009). However, as Matl et al. (2018) note, striving for perfect fairness can inadvertently increase operational costs by requiring more vehicles or longer travel distances, underscoring the need for trade-offs between fairness and efficiency. Methods like the epsilon constraint approach are frequently employed to strike this balance, converting multi-objective problems into single-objective forms by applying secondary constraints (Mavrotas, 2009). Neglecting workload fairness, while potentially reducing costs in the short term, can lead to long-term disruptions. Overburdened drivers or fleets are more likely to withdraw from cooperative agreements, resulting in turnover, route reassignments, and labor disputes that escalate operational expenses (Fernández et al., 2022). This instability threatens fleet cohesion and undermines the sustainability of logistics operations, highlighting the broader implications of inequitable practices. Recent advancements reflect growing interest in this area. Gansterer et al. (2018, 2021) introduced workload fairness models for single-vehicle and multi-vehicle routing problems, later extending their research to multi-depot scenarios. Despite these contributions, their models often lack considerations for time windows and real-world EV constraints, limiting their applicability in emerging logistics networks. Soriano et al. (2023) further

addressed profit fairness in multi-depot vehicle routing problems (MDVRP), advocating for equitable earnings distribution across fleets. Nevertheless, existing models frequently overlook key operational factors such as EV energy consumption, charging costs, and customer time windows, leaving gaps in addressing the practical challenges of sustainable fleet management.

2.4 Exact method, Heuristic, and Metaheuristic Algorithms in Logistics

2.4.1 Exact Method

Exact methods guarantee optimal solutions by systematically exploring the search space while employing techniques to reduce the number of solutions evaluated, ensuring the global optimum is preserved (Amor et al., 2023). The complexity of the LRP arises from the merger of two NP-hard problems facility location and vehicle routing - into a single framework, which requires sophisticated approaches to be developed in order to solve it. To address this, Berger et al. (2007) and Akca et al. (2009) introduced branch-and-price (B&P) algorithms for LRP, while Belenguer et al. (2011) refined the process through branch-and-cut (B&C) with two-index formulations and optimality cuts, enhancing linear relaxation strength. Baldacci et al. (2011) extended this with a three-stage method, reducing facility subsets and solving MDVRP subproblems, significantly tightening bounds. Contardo et al. (2013a, 2013b) expanded on arc-variable formulations, introducing additional valid inequalities and leveraging cut-and-column generation to improve lower bounds and solve larger instances to optimality.

For time constrained LRPs, Ponboon et al. (2016) applied a branch-and-price approach using column generation to solve the master problem with a simplex algorithm, while the subproblem an elementary shortest path problem with resource constraints (ESPPRC)

– ensures convergence by iterating until no columns with negative reduced costs remain. Similarly, Capelle et al. (2019) utilized column generation with a shortest path subproblem for LRPTW, achieving exact solutions for small instances and near-optimal results for larger cases. Paz et al. (2018) tackled the Multi-Depot Electric Vehicle Location Routing Problem with Time Windows (MDEVLRP-TW) using a Mixed Integer Linear Programming (MILP) model, incorporating battery swapping and partial recharging strategies. The model, solved with CPLEX 12.5, minimizes total distance while optimizing vehicle and station allocations.

Recent advancements include multi-objective MILP models solved using GAMS (Hashemi et al., 2022) and two-stage integer programming combined with savings algorithms (Huang et al., 2021). Rabbani et al. (2017) integrated simulated annealing with MILP for blood collection networks, while Leyrer et al. (2020) applied MILP to urban logistics. Despite their ability to guarantee optimality, exact methods often face scalability challenges, necessitating hybrid approaches with metaheuristics to manage larger instances effectively.

2.4.2 Role of Heuristics and Metaheuristics

Metaheuristics play a major role in tackling complex optimization problems—particularly when exact methods become impractical—by delivering approximate yet high-quality solutions within acceptable computational times (Schneider & Drexler, 2017). They are highly adaptable to varying problem structures, large-scale scenarios, and real-world constraints. Early heuristic attempts at solving the Multi-Depot Vehicle Routing Problem (MDVRP) were proposed by Tillman and Cain (1972) and Wren and Holliday (1972). Tillman and Cain (1972) built on Clarke and Wright’s (1964) savings algorithm, while subsequent heuristic extensions introduced considerations such as time windows (MDVRPTW), dynamic control (Su, 1999), split delivery (MDCVRPSD), and heterogeneous fleets—illustrating the progressive complexity of MDVRP research. In more recent work, Soriano et al. (2023) addressed profit fairness in MDVRP using an Adaptive Large Neigh-

neighborhood Search (ALNS) method.

Several other heuristic-based studies underscore the breadth of applications. Reiners (2015), for example, adopted a constraint programming-based heuristic for pre-planned routes. Bae and Moon (2016) extended the MDVRPTW by minimizing total costs tied to depots, vehicles, transportation, and labor. Another significant extension involves integrating location decisions under uncertainty: Zarandi et al. (2011) examined a multi-depot capacitated location-routing problem (MDCLRP) with fuzzy travel times, solved via a simulation-based Simulated Annealing (SA) method. Similarly, Ting and Chen (2013) applied a multi-colony Ant Colony Optimization (ACO) strategy for the Location Routing Problem (LRP), dividing the problem into facility location, customer assignment, and vehicle routing subproblems, each refined by local search.

Hybrid methods also abound. Marinakis and Marinaki (2008b) combined Particle Swarm Optimization (PSO), Greedy Randomized Adaptive Search Procedure (GRASP), Expanding Neighborhood Search (ENS), and Path Relinking (PR) to handle location-routing instances (Perl and B benchmarks), obtaining solutions comparable to Barreto et al. (2007). In parallel, Marinakis and Marinaki (2008a) introduced a bilevel programming formulation for the same problem, solved with GA. Indeed, GAs remain popular in metaheuristics due to their robust exploration capabilities (Goldberg, 1989; Mitchell, 1998). By simulating natural selection, they use selection, crossover, and mutation operators to evolve solutions and avoid local optima (Prins, 2004). Ye and Li (2007) applied a GA to a stochastic LRP by randomly opening depots and then routing vehicles using a savings-based heuristic, demonstrating GAs' adaptability to uncertain settings.

Advances in GA-based research include Lopes et al. (2016), who used variable-length chromosomes to represent both facility locations and vehicle routes, employing route copy crossover, mutation, and local search for refinement. Their approach terminates after a predefined number of non-improving iterations. Yu et al. (2019) proposed a Hybrid Genetic Algorithm (HGA) for the Capacitated Location Routing Problem (CLRP),

merging GAs with local search to achieve a balance between exploration and exploitation. A multi-sequence crossover operator optimized depots and vehicle routes simultaneously, outperforming benchmark results in the literature.

Hybridizing GAs with other metaheuristics can yield even stronger outcomes. According to Talbi (2002), GA-ACO hybrid frameworks combine the GA's broad search scope in facility selection and vehicle assignment (Prins, 2009) with ACO's efficient path construction, especially for combinatorial subproblems like routing. Pareto-based optimizations further enhance these multi-criteria approaches (Doerner et al., 2006). By harnessing algorithmic synergies, hybrid methods often surpass single-algorithm counterparts in solution quality and computational speed (Stodola, 2020). The use of hybrid algorithms is especially beneficial in time-sensitive VRPs, as demonstrated by Reimann et al. (2004), which show improvements in both solution quality and runtime for VRP instances with time windows.

Lastly, recent studies have adapted hybrid algorithms to specific scenarios, including humanitarian logistics. Wei et al. (2020) employed a hybrid ACO and particle swarm approach to optimize depot locations and vehicle routes for post-disaster relief, minimizing costs and time-window penalties. Tadaros et al. (2020) combined GAs and ACO to address the LRP, improving solution diversity and convergence rates. This method exploits the global exploratory nature of GAs and ACO's ability to handle complex pathfinding, leading to a more robust balance of exploration and exploitation. Thus, the growing body of research confirms that combining multiple metaheuristics is a promising direction for tackling intricate, large-scale location-routing problems across diverse industries.

Table 2.1: Literature Review Comparison Table

Papers	VRP	EVRP	FLP	LRP	MDVRP	MDFLP	MDLRP	Time Window	Bi-objective	Fairness	Exact Method	Metaheuristic	Heuristic
Feng & Figliozzi, 2013	✓	✓	×	×	×	×	×	×	×	×	✓	×	×
Davis & Figliozzi, 2013	✓	✓	×	×	×	×	×	×	×	×	✓	×	×
Kyriakakis et al., 2022	✓	✓	×	×	×	×	×	×	✓	×	×	✓	✓
Schneider et al., 2014	✓	✓	×	×	×	×	✓	✓	✓	×	✓	✓	×
Zang et al., 2022	✓	✓	×	×	×	×	×	✓	×	×	✓	×	×
Gnann et al., 2018	×	✓	✓	✓	×	×	×	×	✓	×	×	×	✓
Conrad & Figliozzi, 2011	✓	✓	×	×	×	×	×	✓	×	×	×	×	✓
Shao et al., 2018	✓	✓	×	×	×	×	×	×	×	×	×	✓	×
Hiermann et al., 2016	✓	✓	×	×	×	×	×	✓	×	×	✓	✓	×
Desaulniers et al., 2016	✓	✓	×	×	×	×	×	✓	×	×	✓	×	×
Felipe et al., 2014	✓	✓	×	×	×	×	×	×	×	×	×	✓	✓
Keskin & Çatay, 2018	✓	✓	×	×	×	×	×	✓	×	×	✓	✓	✓
Keskin & Çatay, 2016	✓	✓	×	×	×	×	×	✓	×	×	×	✓	×
Xiao et al., 2012	✓	×	×	×	×	×	×	×	×	×	×	✓	×
Salhi and Nagy, 1999	×	×	✓	✓	×	×	×	×	×	×	×	×	✓
Averbakh and Berman 1994.	✓	×	✓	✓	×	×	×	×	×	×	✓	×	✓
Desaulniers et al., 2002	✓	×	×	×	✓	×	×	✓	×	×	×	✓	×
Baldacci et al., 2012	✓	×	×	×	×	×	×	✓	×	×	✓	×	×
Worley et al., 2012	✓	✓	×	✓	×	×	×	×	×	×	✓	×	×
Schiffer and Walther 2017	✓	✓	×	✓	×	×	×	✓	✓	×	✓	×	×
Schiffer and Walther 2018	✓	✓	✓	✓	×	×	✓	✓	×	×	×	✓	×
Yang and Sun 2015	✓	✓	×	✓	×	×	×	×	✓	×	✓	✓	×
Cordeau et al., 1997	✓	×	×	×	✓	×	×	×	×	×	×	✓	×
Capelle et al., 2019	✓	×	✓	✓	×	×	×	✓	×	×	✓	×	×
Paz et al., 2018	✓	✓	✓	✓	✓	✓	✓	✓	×	×	✓	×	×

Papers	VRP	EVRP	FLP	LRP	MDVRP	MDFLP	MDLRP	Time Window	Bi-objective	Fairness	Exact Method	Metaheuristic	Heuristic
Kallehauge et al., 2005	✓	×	×	×	×	×	×	✓	×	×	✓	×	×
Parayoga et al., 2022	✓	×	✓	✓	×	×	×	✓	✓	×	×	✓	×
Xiang et al., 2008	✓	×	×	×	×	×	×	✓	×	×	×	✓	×
Manchini et al., 2021	✓	×	×	×	✓	×	✓	✓	✓	✓	✓	✓	×
Hezarkhani et al., 2016	×	×	×	×	×	×	×	×	×	✓	×	×	×
Koç and Bektaş 2014	✓	×	×	×	×	×	×	✓	✓	✓	×	✓	×
Lim and Zhang 2007	✓	×	×	×	×	×	×	✓	✓	✓	✓	×	×
Jozefowiec et al 2009	✓	×	×	×	×	×	×	×	✓	✓	×	✓	×
Bao et al 2021	×	✓	✓	✓	×	×	×	×	✓	×	✓	×	×
Wang et al., 2023	×	×	✓	✓	×	×	×	×	✓	×	✓	×	✓
Du et al., 2023	✓	✓	✓	✓	✓	✓	✓	✓	✓	×	×	✓	×
Portela et al., 2021	✓	✓	✓	✓	✓	×	✓	✓	×	×	×	✓	×
Gansterer et al., 2021	✓	×	✓	✓	×	×	×	✓	✓	✓	×	×	✓
Soriano et al., 2023	✓	×	×	×	✓	×	×	×	✓	✓	×	✓	×
Our Research	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	×	✓	×

2.5 Gap Analysis and Research Opportunities

2.5.1 Identified Research Gaps

In the literature review (Table 2.1), we present a comprehensive overview of the evolving research landscape on EVRP, LRP, MDLRP, and workload fairness. Our study addresses critical gaps by integrating multiple objectives and constraints into a unified model, setting it apart from prior work. While numerous studies have independently explored VRP and EVRP, fewer have incorporated FLP and LRP within a multi-depot, time-constrained framework specifically designed for EVs.

Unlike existing models that prioritize cost minimization, our research places workload fairness at the forefront, ensuring equitable distribution of delivery tasks. This approach not only enhances operational efficiency but also mitigates labor unrest, as high employee satisfaction reduces the likelihood of strikes (Asogwa & Nosike, 2024). By embedding fairness constraints directly into the MDLRP structure, our work expands on contributions from Gansterer et al. (2021) and Soriano et al. (2023), addressing real-world challenges such as driver fatigue and turnover. Additionally, we introduce a metaheuristic algorithm that hybridizes GA with ACO to tackle large-scale problems effectively. While studies like Paz et al. (2018) and Schiffer & Walther (2017) explore depot location and EV routing, our model extends this by incorporating depot capacity, energy consumption, and vehicle recharging constraints. This dual focus on cost reduction and workload fairness represents a significant advancement in sustainable urban logistics.

Our research bridges the divide between theoretical optimization and practical logistics applications, contributing to more sustainable urban delivery systems. Although existing studies provide valuable VRP and FLP models, practical validation remains limited. Unbalanced workloads and poor conditions often lead to labor disputes, highlighting the importance of fairness in preventing strikes (Time, 2024). By integrating multi-depot location routing with time-window constraints and workload fairness into a scalable,

comprehensive framework for EVs, our study directly addresses this gap, advancing both logistical efficiency and employee well-being.

Chapter 3

Methodology

3.1 Introduction

This section details a mathematical model designed to solve a Bi-Objective MDLRPTW for EVs, integrating objectives for cost minimization and workload fairness. Our model is formulated as a Mixed Integer Linear Programming (MILP) problem, capturing the complexity of EV-based logistics with a linear structure. We implemented a model that includes several constraints related to vehicle capacity, depot assignment, customer service time windows, and the requirements of the unique infrastructure for EV charging. This approach is well suited for modern logistics networks seeking to reduce costs while at the same time achieving a balanced workload for drivers as an important goal.

3.2 Model Notation

In the following section, we provide the model notation used to formulate the proposed Bi-Objective MDLRPTW model. This notation includes all sets, indices, parameters, and decision variables, offering a comprehensive reference for understanding the mathematical formulation.

The indices i and j represent locations in the network, where i is the origin (starting

location) and j is the destination. These indices are used to specify any location within the network, which includes both depots and customer sites, denoted as set V . The indices can be used interchangeably to track the flow between any two locations, whether it is from a depot to a customer, a customer to a depot, or even customer-to-customer. The symbol o represents a depot location and refers to a specific depot within the set of all possible depot locations in the model. This helps to distinguish depot locations from other types of locations. The set u denotes the customer set, which specifically includes only customer locations and excludes depots. This distinction ensures that any customer-specific parameters and constraints are applied only to designated customer locations, avoiding any mix-up with depot-related considerations. The set K defines the EVs which helps to understand vehicle movements within the network. The notation structure is created so that the relationships and constraints within the model can be clearly defined, thus assisting in maintaining a clear differentiation between depots and customers as well as tracking the flow of routes between them in a consistent manner. Each element is defined in the tables below.

Sets	Definitions
V	All locations, including customers and potential depots.
K	Electric vehicles.
o	Depot locations.
u	Customers.

Table 3.1: Definitions of Sets

Notation	Definitions
A_o	Depot capacity.
DE_j	Customer demand.
Q_k	Vehicle capacity of EVs.
Vc	Vehicle fixed cost (Driver's salary for one year).
yc	Fixed number of working days in year
S_i	Service time for each customer.
$T_{start,i}$	Earliest time window for each customer.
$T_{end,i}$	Latest time window for each customer.
T_{max}	Maximum allowable time for last-time delivery.
D_{ij}	Distance between locations.
T_{ij}	Travel time between locations based on speed.
p	Depot installation cost with charging piles.
elc	Electricity price in Ontario.
e_{rate}	Energy consumption rate for EVs.
ϵ	Vehicle total kilometers.
$Speed$	Average vehicle speed.
n	Optional maximum number of depots selected from potential depot locations.

Table 3.2: Definitions of Parameters

Variables	Definitions
z_{ijk}	Binary variable equals 1 if vehicle k travels between location i to j , otherwise 0.
σ_k	Binary variable equals 1 if vehicle k is used, otherwise 0.
y_o	Binary variable equals 1 if depot o is selected, otherwise 0.
γ_{ok}	Binary variable equals 1 if vehicle k is assigned to depot o , otherwise 0.
r_{ok}	Binary variable equals 1 if customer j is assigned to depot o , otherwise 0.
w_{ojk}	Auxiliary binary variable for linking depot, vehicle, and customer assignment.
ϕ_{oijk}	Auxiliary binary variable ensures vehicles travel between customers assigned to the same depot.
X_i	Positive variable for Arrival at time at location i
λ	Maximum distance traveled by any vehicle

Table 3.3: Definitions of Variables

3.3 Mathematical Model

Our approach to the problem is an optimization model, which is based on two main components, the objective function and the constraint. The objective functions define the goals of the optimization process, typically aiming to either minimize or maximize a particular measure of interest, while the constraints impose limitations to ensure that feasible solutions adhere to real-world restrictions (Winston & Venkataramanan, 2003). Our model considers two primary objectives: total cost minimization and workload fairness.

3.3.1 Objective Functions

We calculate the objective function based on sets, notations, and variables in table 3.1, table 3.2, and table 3.3 respectively :

Total Cost Objective Function (1)

$$\text{Min} \left(\sum_{ijk} d_{ij} \cdot z_{ijk} \cdot e_{\text{rate}} \cdot elc \cdot yc + vc \sum_k \sigma_k + p \sum_o y_o \right)$$

Workload Fairness Objective Function (2)

$$\text{Minimize Max Distance} \sum_{ij} d_{ij} \cdot z_{ijk} \leq \lambda \rightarrow \lambda \leq \epsilon$$

In our total cost objective function(1), the first Term represents the travel cost. Here d_{ij} is the distance between location i and j , multiplied by z_{ijk} , a binary variable $[0, 1]$ that indicates whether vehicle k travels between location i and j , equal to 1 if it does, otherwise 0. This term is summed across all locations and vehicles, calculating the travel distance, which is then multiplied by e_{rate} , the energy consumption rate for EVs. This allows us to determine how much energy each vehicle consumes while traveling between locations. Finally, by multiplying elc , the electricity price in Ontario, we can determine the total travel cost for each vehicle across all trips in one year if we multiply by yc which is a fixed number of 365 days.

The second term represents the objective function is the cost of vehicle usage. Here vc is the fixed driver's cost which is multiplied by σ_k , a binary variable $[0, 1]$ that indicates whether vehicle k is used (1 if used, 0 otherwise). This term is summed over all vehicles to capture the total cost of vehicle usage.

The third term of the objective function represents the cost of the installation of the depot. Here, p is the depot installation cost, which is multiplied by y_o , a binary variable $[0, 1]$ that indicates whether the depot is installed (1 if installed, 0 otherwise). Summing this term overall potential depots gives the total installation cost of the depot. By minimizing the objective function (1), we can determine the total cost in the model, considering travel costs, vehicle usage, and depot installation.

The second objective function(2) focuses on workload fairness between all drivers by minimizing the maximum travel distance for each vehicle k across all trips. This objective is handled using the ϵ -constraint methods, first introduced by Laumanns et al. in 2006, proposing an alternative method for solving problems by optimizing one objective while limiting the others in constraints. We aim to keep each vehicle's total travel distance below a threshold λ , with $\lambda \leq \epsilon$. Here, ϵ represents the maximum distance each vehicle can travel based on its battery capacity.

3.3.2 Constraints

In this section, we present a set of constraints that govern a feasible solution to the optimization problem in the context of the optimization model. These constraints ensure that the solution satisfies all real-world restrictions, such as vehicle constraints, time windows, depot constraints, and customer constraints in linear, while also maintaining the balance between the objectives of cost minimization and workload fairness.

Vehicle capacity constraint (3)

$$\sum_{ij} DE_j \cdot z_{ijk} \leq Q_k \cdot \sigma_k \quad \forall k$$

Constraint(3) ensures each vehicle does not exceed its carrying capacity when serving assigned customers. For each vehicle k , the left side calculates the total demand DE_j that vehicle k is set to meet, considering all customer locations it will visit. The right side represents the vehicle's capacity Q_k if the binary variable $\sigma_k = 1$ (meaning the vehicle is in use) or 0 if $\sigma_k = 0$ (meaning the vehicle is not in use).

Vehicle flow constraint (4)

$$\sum_i z_{ijk} = \sum_i z_{jik} \quad \forall j \in u$$

Constraint(4) is a balance constraint that ensures the number of vehicles entering must equal the number of vehicles leaving for locations. This prevents vehicles from getting stuck at any location.

Enforce customer visit (5)

$$\sum_{k,i} z_{iuk} = 1 \quad \forall j \in u$$

Constraint(5) forces each customer to be visited exactly once, ensuring each demand point is serviced.

Subtour Elimination (6)

$$z_{ijk} = 0 \quad \forall i = j$$

Constraint(6) by enforcing this, the model prevents vehicles from looping back to locations they have already visited, thereby maintaining a continuous.

Depot selection constraint (7)

$$\sum_o y_o \leq n$$

Constraint(7) limits the number of depots that can be installed based on the maximum

number of depots allowed from potential depot locations specified.

Depot Capacity Constraint (8)

$$\sum_{j \in U} DE_j \cdot r_{oj} \leq A_o \cdot y_o \quad \forall o \in V$$

Constraint(8) ensures that when $y_o = 1$ (meaning depot o is active) the sum of demands (DE_j) for all customers assigned to this depot which is weighted by binary variable r_{oj} does not exceed depot's capacity A_o . If $y_o = 0$ (meaning the depot is not opened), the right side becomes 0, forcing the left side to be 0 as well. This makes sure that no customers are assigned to a depot that is not open.

Depot Assignment to Vehicles (9)

$$\gamma_{ok} \leq y_o \quad \forall o \in V, \forall k \in K \quad (9.1)$$

$$\sum_{k \in K} \gamma_{ok} \geq y_o \quad \forall o \in V \quad (9.2)$$

Constraint(9) this constraint has two parts, the first part(9.1) ensures that vehicles can only be assigned to depots that are actually selected from potential depot locations. If a depot is not selected ($y_o = 0$), then $\gamma_{ok} \leq 0$ forces γ_{ok} to be 0 for all vehicles k . This means no vehicles can be assigned to depot o if it is not active. The second part(9.2) guarantees that every selected depot(where $y_o = 1$) has at least one vehicle assigned to it by requiring that the summation of γ_{ok} over all vehicles is at least 1.

Vehicle Assignment to Exactly one Depot (10)

$$\sum_{o \in V} \gamma_{ok} = \sigma_k \quad \forall k \in K$$

Constraint(10) ensures that each vehicle is assigned to only one depot if it is in use.

This constraint maintains an organized route structure. Additionally, it restricts unused vehicles when $\sigma_k = 0$ (indicating a vehicle is not in use), it is not assigned to any depot.

Customer Depot Assignment (11)

$$\sum_o r_{oj} = 1 \quad (11.1)$$

$$r_{oj} \leq y_o \quad (11.2)$$

Constraint(11) has two parts, In (11.1) ensures each customer is assigned to only one depot by requiring that $r_{oj} = 1$ where r_{oj} is a binary variable set to 1 if customer j is assigned to depot o , and 0 otherwise. In (11.2) y_o is a binary variable indicating whether depot o is open (1 if open, 0 if closed). If $y_o = 0$, then r_{oj} must also be 0, meaning customer j cannot be assigned to a closed depot. Together, these ensure each customer is linked to a single opened depot, supporting an organized assignment structure.

Time starts at zero in depot (12)

$$X_o = 0 \quad \forall o$$

Constraint(12) ensures all starting time at each depot is equal to zero, which serves as a reference point for all vehicle routes.

Time Windows (13)

$$T_{\text{start},i} \leq X_i \leq T_{\text{end},i} \quad \forall j \in u$$

Constraint(13) enforces the time window for each customer. $T_{\text{start},i}$ is the earliest time a

customer can be served, while $T_{\text{end},i}$ is the latest time allowable service time.

Subtour elimination constraint & Tracking time (14)

$$z_{ijk} = 1 \rightarrow X_j \geq X_i + s_i + T_{ij} \quad \forall i, j \in V : i \neq j, j \neq 0, \forall k \in K \quad (14.1)$$

$$z_{io k} = 1 \rightarrow X_i + s_i + T_{io} \leq T_{\max} \quad \forall i \in I, \forall k \in K \quad (14.2)$$

Constraint(14.1) applies when a vehicle k travels from location i to location j (indicated by binary variable $z_{ijk} = 1$). It ensures that arrival time X_j at location j is at least the sum of the departure time from location i X_i , the service time at i (s_i), and the travel time between i and j (T_{ij}). As a result of this requirement, the sequence of visits remains valid and subtours can be avoided by requiring each stop to progress forward in time without returning to previously visited locations.

Constraint(14.2) applies when a vehicle k returns from a location i to its assigned depot o (indicated by binary variable $z_{io k}=1$). It ensures that the vehicle arrives back at the depot within the maximum allowable time $T_{\max}$. Here, $X_i + s_i + T_{io}$ represents the time required to complete service at i and travel back to the depot. As a result, this constraint ensures that routes end within the maximum time window. Since this constraint is nonlinear, we need to apply linearization methods to transform it into a linear form, making it compatible with the model's requirements.

Linearization of Tracking time & Time Window (15)

$$X_j \geq X_i + s_i + T_{ij} \cdot z_{ijk} - T_{\max} \cdot (1 - z_{ijk}), \quad \forall k, \forall i, \forall j, \quad u(j) \text{ and } i \neq j \quad (15.1)$$

$$X_i + (s_i + T_{io}) \cdot z_{io k} \leq T_{\max} + T_{\max} \cdot (1 - z_{io k}), \quad \forall k, \forall i, \forall o, \quad u(i) \text{ and } o(o) \quad (15.2)$$

Constraint (15) is a linear form of constraint (14) which enforces the time feasibility for any vehicle traveling from customer i to customer j . Constraint(15.1) is a term that is only applied when the route is active ($z_{ijk} = 1$), the last term, $-T_{max} \cdot (1 - z_{ijk})$ deactivates the constraint when ($z_{ijk} = 0$). Constraint (15.2) ensures that any vehicle k returning from customer i to its assigned depot o completes its route within the maximum allowable time (T_{max}). In this term, the binary $z_{io k} = 1$ indicates whether vehicle k travels from customer i to depot o otherwise if $z_{io k} = 0$ the last term ($-T_{max} \cdot (1 - z_{io k})$) adds sufficient slack to deactivate the constraint, ensuring that it does not interfere with other feasible solutions.

Vehicle start and Return from Assigned depot (16)

$$\sum_{j \in U} z_{ojk} \leq \gamma_{ok} \quad \forall o \in V, k \in K \quad (16.1)$$

$$\sum_{i \in V} z_{io k} \leq \gamma_{ok} \quad \forall o \in V, k \in K \quad (16.2)$$

Constraint(16) makes sure that each vehicle assigned to a depot ends and starts its route at that depot and prevents vehicles from starting or ending at unassigned depots. This constraint has two parts. Constraint(16.1) ensures that if a vehicle k is assigned to depot o (indicated by $\gamma_{ok} = 1$), it can depart from depot o to serve customers. Specifically, the sum $\sum_{j \in u} z_{ojk}$ represents all possible trips from depot o to any customer j . This sum is constraint by γ_{ok} , which means that if $\gamma_{ok} = 1$, the vehicle can depart from depot o ; if $\gamma_{ok} = 0$, the vehicle cannot leave from that depot.

Constraint(16.2) ensures that each vehicle k assigned to a depot o also returns to that same depot by the end of its route. Here, $\sum_{i \in V} z_{io k}$ indicates all trips that end at depot o from any other location i . This sum is similarly constrained by γ_{ok} ensuring that the vehicle only returns to depot o if it was originally assigned to that depot.

Depot-Vehicle-Customer Link (17)

$$z_{ojk} \leq \gamma_{ok} \cdot r_{oj} \quad \forall o \in V, j \in u, k \in K$$

Constraint (17) requires the vehicle and customer to be assigned to the same depot before it can serve a customer. As a result of this constraint, depots, vehicles, and customers are logically linked. In this constraint(17) z_{ojk} is a binary variable indicating whether vehicle k travels from depot o to customer j . Here, $\gamma_{ok} = 1$ if vehicle k is assigned to depot o , and $r_{oj} = 1$ if customer j is assigned to depot o . If both γ_{ok} and r_{oj} equal 1, z_{ojk} can be 1 to allowing vehicle k to serve customer j from depot o . However, if either γ_{ok} and r_{oj} are 0 then z_{ojk} must be 0, meaning that the vehicle k cannot serve the customer unless both are assigned to the same depot.

Since this constraint(17) involves multiplying two binary variables (γ_{ok}, r_{oj}) , it is non-linear, so we need to apply the linearization method to make it compatible with linear optimization. We will address this in the next constraint.

Linearization of Depot-Vehicle-Customer Link (18)

$$z_{ojk} \leq w_{ojk} \quad (18.1)$$

Constraint (18.1) ensures that if $z_{ojk} = 1$ (meaning vehicle k travels from depot o to customer j), then w_{ojk} must also be 1 which is an auxiliary variable.

$$\sum_k w_{ojk} = r_{oj} \quad \forall o, j \quad (18.2)$$

Constraint (18.2) ensures that the sum of w_{ojk} across all vehicle k is equal to r_{oj} , which indicates whether customer j is assigned to depot o . In other words, if $r_{oj} = 1$ (customer j is assigned to depot o), then at least one w_{ojk} must be 1 for some vehicle k . If $r_{oj} = 0$

(customer j is not assigned to depot o), then all w_{ojk} for that depot-customer pair must be 0.

$$w_{ojk} \leq \gamma_{ok} \quad (18.3)$$

Constraint (18.3) ensures that w_{ojk} can only be 1 if vehicle k is assigned to depot o when $\gamma_{ok} = 1$. If $\gamma_{ok} = 0$ then w_{ojk} must also be 0, meaning vehicle k cannot serve customer j from depot o .

The purpose of these constraints is to linearize constraint (17).

Customer to Customer Link (19)

$$z_{ijk} \leq \gamma_{ok} \cdot r_{oi} \cdot r_{oj} \quad \forall i, j \in u, o \in V, k \in K \quad (19.1)$$

Constraint (19) ensures the vehicle can travel between customers only if both customers are assigned to the same depot and the vehicle is assigned to that depot as well. This works as follows, if both customers i and j are assigned to depot o ($r_{oi} = 1, r_{oj} = 1$) and vehicle k is also assigned to depot o ($\gamma_{ok} = 1$) then $\gamma_{ok}, r_{oi}, r_{oj} = 1$. Following that, $z_{ijk} \leq 1$ which allows z_{ijk} to be 1 if vehicle k indeed travels from customer i to customer j . However, if either customer i and j is not assigned to depot o or vehicle k is not assigned to depot o , the right side of the equation becomes 0, so this forces $z_{ijk} = 0$, meaning that vehicle k cannot travel. Since this constraint involves multiplying three binary variables (γ_{ok}, r_{oi} , and r_{oj}) on the right side, it is nonlinear. Linearization will be applied to the next constraint to ensure it is compatible with the model's linear optimization requirements.

Linearization of Customer to Customer Link (20)

$$\phi_{oijk} \leq r_{oi} \quad \forall i \in u \quad (20.1)$$

$$\phi_{oijk} \leq r_{oj} \quad \forall j \in u \quad (20.2)$$

$$\phi_{oijk} \leq \gamma_{ok} \quad (20.3)$$

$$\phi_{oijk} \geq r_{oi} + r_{oj} + \gamma_{ok} - 2 \quad (20.4)$$

Constraint (20) is a linearization of the method that is applied to constraint (19). To achieve this, we introduce an auxiliary binary variable, ϕ_{oijk} which allows us to express the original constraint without directly multiplying the binary variables. We split these three binary variables r_{oi} , r_{oj} , and y_{ok} into separate constraints. In Constraint (20.1) we ensure ϕ_{oijk} can only be 1 if $r_{oj} = 1$ meaning customer i is assigned to depot o . In constraint (20.2) we ensure that ϕ_{oijk} can only be 1 if $r_{oj} = 1$, meaning customer j is assigned to depot o . In constraint (20.3) we ensures that ϕ_{oijk} can only be 1 if $y_{ok} = 1$ meaning vehicle k is assigned to depot o . So, right now we can add them together in constraint (20.4), we ensure that $\phi_{oijk} = 1$ only when these three variables r_{oi} , r_{oj} , and y_{ok} are 1. If any of these variables is 0, the right side of the inequality will be less than 1, forcing ϕ_{oijk} to be 0.

Chapter 4

Solution Algorithm

To address the Bi-Objective MDLRPTW, we developed a mathematical model formulated as a MILP model in GAMS. MILP is a powerful tool for optimization in logistics and transportation due to its ability to handle both discrete and continuous decision variables, accommodating complex constraints such as vehicle capacities, depot assignments, time windows, and EV constraints. GAMS provides a robust environment for implementing MILP models, offering precise control over variables, constraints, and objective functions. By formulating the MDLRP as a MILP, we leverage exact optimization methods to find the mathematically optimal solution within the defined constraints with the CPLEX solver. In this method, we can find an exact solution that is the best possible answer to an optimization problem, considering all constraints and the objective function. Solving for an exact solution can be very computationally intensive, especially for large or complex problems.

MIP formulations become computationally infeasible for large-scale MDLRP instances. The time and memory required to solve the MIP model to optimality increase exponentially with the problem size (Cordeau et al., 2007).

Being a non-deterministic polynomial problem or NP-Hard problem means that up until now, there has been no efficient solution to this problem. This is the case for VRP,

which uses heuristic or metaheuristic algorithms to find an approximate solution if the number of customers and constraints is high. The purpose behind it is to look for the best and optimal way to reach customers from a starting point (Depot) and under many constraints (Laporte, 1992). An NP-hard problem is one that is at least as hard as the hardest problems in NP (nondeterministic polynomial time). These problems do not have known algorithms that can solve them in polynomial time, and finding an exact solution becomes computationally infeasible as the problem size increases (Garey & Johnson, 1979). The VRP involves finding the optimal set of routes for a fleet of vehicles to deliver goods to a set of customers which is known as the NP-Hardness problem (Lenstra & Rinnooy Kan, 1981). Additionally, the FLP involves selecting optimal locations for facilities to minimize costs associated with opening facilities and serving customers. It is proven to be NP-hard because it includes the set covering problem and the p-median problem as special cases, both of which are NP-hard (Krarup & Pruzan, 1983). The LRP is considered NP-hard due to its inherent computational complexity, which arises from the combination of two classic optimization problems: FLP and VRP. In fact, FLP and VRP both have themselves been proven to be NP-hard problems, and the LRP is arguably even more challenging to solve because it incorporates both of these challenges at once (Nagy & Salhi, 2007).

Since MDLRP is an extension of the standard LRP that includes multiple depots into decision-making which makes the problem more complex. Decisions about which depots to open from potential locations, how to assign customers to depots, and how to route vehicles to customers. These problems add layers of complexity beyond those standard LRP that make MDLRP NP-Hardness a problem.

Since exact algorithms are impractical for large NP-hard problems, heuristics and metaheuristics provide approximate solutions in a reasonable timeframe (Osman & Laporte, 1996). Metaheuristic algorithms make a balance between time and quality. It means that metaheuristic algorithms are designed to find high-quality solutions that are close

to optimal within acceptable computational times(Blum & Roli, 2003). Metaheuristic algorithms are capable of handling large-scale problems that are intractable for exact methods and easily adaptable to various problem-specific constraints and objectives, such as time windows, multiple vehicle types, and stochastic demands(Talbi, 2009)(Gendreau & Potvin, 2010).

These algorithms are designed to explore the solution space efficiently by combining different concepts for exploring and exploiting the search space to find near-optimal solutions for complex optimization problems (Blum & Roli, 2003). There are several type of metaheuristic algorithms like Genetic, Ant Colony, Tabu search, etc. An example of a metaheuristic algorithm is the GA, which is inspired by the biological evolution process(Michalewicz & Schoenauer 1996). GA is a search heuristic inspired by Charles Darwin’s theory of natural evolution. It reflects the process of natural selection where the fittest individuals are selected for reproduction to produce offspring of the next generation (Holland, 1992).

The basic elements of GA are chromosome representation, fitness selection, and biological-inspired operators. Additionally, Holland introduced a novel element in the form of inversion that is generally used in GA implementations(Holland JH 1975). It is common for chromosomes to be formatted as binary strings. In chromosomes, each locus (specific location on the chromosome) has two alleles (different forms of genes). In the solution space, chromosomes are considered points. These are processed using genetic operators by iteratively replacing its population. (Katoch et al 2021). Using the fitness function, one can assign values to all the chromosomes in a population and calculate their fitness(Michalewicz & Schoenauer 1996). The biologically inspired operators are selection, mutation, and crossover. As part of selection, chromosomes are selected based on their fitness value. In crossover operators, subsequences between chromosomes are changed at a random locus to create offspring. As a result of mutation, some bits of the chromosome are randomly flipped based on probability, leading to a new chromosome (Michalewicz

& Schoenauer, 1996; Michalewicz, 1992; Holland, 1975).

ACO is a metaheuristic inspired by the foraging behavior of ants seeking paths between their colony and food sources. This optimization method was first introduced by Dorigo and colleagues in the early 1990s (Dorigo & Di Caro, 1999). In nature, ants find the shortest paths between their nest and food sources using pheromone trails. A chemical substance called pheromone is deposited on the ground as ants travel. Other ants sense these pheromones and are more likely to follow paths with stronger pheromone concentrations. Shorter paths accumulate pheromones more quickly because ants traverse them more frequently, leading the colony to converge on the most efficient routes (Goss et al., 1989). Additionally, pheromones evaporate over time, preventing the indefinite accumulation of pheromones on any one path and allowing the colony to adapt if better paths become available (Deneubourg et al., 1990)

The ACO algorithm simulates the behavior of real ants by using artificial "ants" that explore a graph representing the problem to find optimal solutions. In this algorithm, artificial ants build solutions gradually by following pheromone trails, numerical values associated with solution components and using heuristic information, such as the inverse of distance in routing problems, to assess the desirability of certain paths. Movement decisions are governed by a probabilistic transition rule that combines pheromone levels and heuristic data, allowing ants to balance between exploiting known good paths and exploring new ones (Dorigo & Gambardella, 1997). Upon completion of a solution, ants update their pheromone levels to reinforce their successful solutions. Pheromone evaporation reduces the strength of all pheromone trails over time, preventing premature convergence on suboptimal solutions and encouraging continuous exploration of the solution space (Dorigo & Di Caro, 1999).

The inter-route exchange is one of the fundamental heuristic techniques used in solving the VRP, which is a factor when it comes to optimizing logistics and transportation networks. IRE was first introduced as part of the foundational strategies in Tabu Search

and was influenced by earlier local search methodologies proposed (Glover, 1989; Fisher & Jaikumar 1981). This method involves the strategic swapping of customers between different vehicle routes with the aim of enhancing overall solution quality. By reassigning customers, IRE seeks to minimize total travel distance and ensure adherence to operational constraints such as time windows and vehicle capacities. This method involves the strategic swapping of customers between different vehicle routes with the objective of enhancing overall solution quality.

GA can be easily hybridized with other optimization methods for improving their performance like ACO, Tabu Search, and many more. This makes the algorithm find better solutions. Hybrid GA provides better solution quality, higher efficiency, a guarantee of feasible solutions, and optimized control parameters over other methods (El-Mihoub et al, 2006).

We made a hybrid metaheuristic algorithm by a combination of GA and ACO to solve the mathematical model on a large scale due to the complexity that it has and GAMS can not solve in an appropriate time. In this model, GA will Handle the higher-level optimization of depot selections, vehicle assignments, and initial route constructions. GA explores a wide range of solutions to avoid local optima. On the other hand, ACO will serve as a local search method within the GA to optimize the routes of individual vehicles. Also ACO fine-tunes individual routes for better efficiency. The combination of these two enhances both exploration and exploitation, leading to higher-quality solutions.

4.1 Flow Chart

There are many steps involved in HGA and we describe each step in a flow chart in general, and then we explain each step in detail to understand how the model works from start to finish.

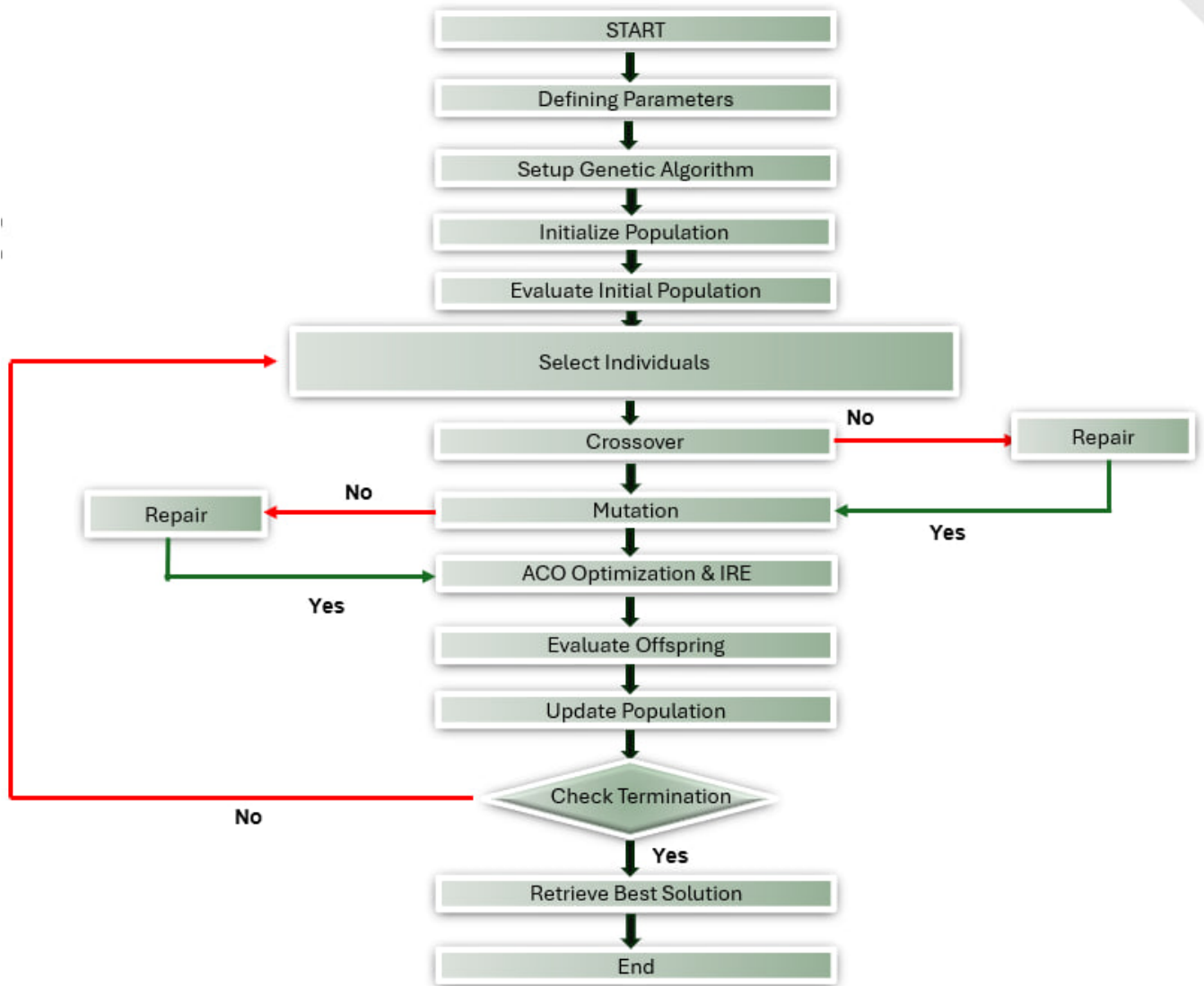


Figure 4.1: Flow chart of Hybrid Genetic Algorithm

4.2 Chromosome Representation

In GA, the chromosome (or individual) serves as the fundamental unit representing a potential solution to the MDLRPTW. During the Setup GA step, each chromosome encapsulates multiple decision variables important for effectively addressing this complex problem. Specifically, in this implementation, the chromosome is structured into four components: the Depot Selection Vector, Vehicle Assignment List, Customer Assign-

ment Dictionary, and Vehicle Routes Dictionary. The depot selection vector is a binary array where each element indicates whether a depot is open (1) or closed (0). This vector plays a key role in determining the operational footprint and associated fixed costs of maintaining depots. The vehicle assignment list assigns each vehicle to a specific depot by index, establishing the starting point for each vehicle's route and impacting route feasibility, service coverage, and load balancing.

The customer assignment dictionary maps each customer to a vehicle, ensuring that all customer demands are met within vehicle capacity limits while also adhering to designated time windows. Lastly, the vehicle routes dictionary specifies the sequence in which each vehicle visits its assigned customers, effectively outlining the route plan. This sequencing is essential for minimizing travel distances and ensuring that all customers are serviced within their designated time windows.

This composite representation enables the GA to manipulate multiple aspects of the solution simultaneously, facilitating a comprehensive exploration of the solution space. By encoding depot selections, vehicle allocations, customer assignments, and route planning within a single chromosome, the algorithm can efficiently navigate the complex interdependencies and constraints inherent in the MDLRPTW, enhancing solution quality and coherence.

Composite chromosome representations, as noted in the literature, are particularly effective for multi-faceted problems like MDLRPTW because they encapsulate various decision layers within a single individual, enabling the simultaneous optimization of multiple components (Prins, 2004; Gendreau et al., 2008). This holistic approach allows genetic operators to explore and exploit the interdependencies between depot selections, vehicle allocations, and routing plans, resulting in high-quality solutions (Bräysy et al., 2004). Moreover, such representations align well with the modular structure of MDLRPTW, where decisions at one level (e.g., depot selection) directly impact other levels (e.g., vehicle routing) (Laporte et al., 2000).

4.3 Population Initialization

Population initialization is the foundational step in the GA, where an initial set of diverse and feasible solutions (chromosomes) is generated. The quality and diversity of this initial population greatly influence the algorithm's convergence speed and the quality of final solutions. In this implementation for the MDLRPTW, the initialization process includes several tailored sub-steps that ensure a well-distributed starting population while embedding problem-specific knowledge.

The first step, depot initialization, begins with random selection, where a subset of depots is randomly chosen to be open, ensuring that the count does not exceed the predefined maximum number. This randomness introduces diversity, preventing bias toward specific depot configurations. The selection is then binary encoded within the chromosome, creating a vector that facilitates genetic operations like crossover and mutation, which can effectively manipulate depot statuses as the algorithm progresses.

Following depot selection, the vehicle assignment process involves random allocation of each vehicle to one of the open depots, which establishes each vehicle's starting point and impacts route feasibility, service coverage, and load balancing. To maintain operational coverage, the algorithm ensures that each open depot has at least one assigned vehicle, preventing scenarios where a depot is open but has no servicing vehicles. Ensuring that each depot has at least one vehicle assigned prevents operational inefficiencies and aligns with real-world logistics practices (Bräysy et al., 2001).

The next stage, customer assignment and route construction employs a proximity-based assignment heuristic to allocate customers to their nearest open depot based on a distance matrix. This approach reduces initial travel distances and balances the workload across depots, enhancing efficiency and load distribution. Route construction then proceeds for each vehicle by constructing feasible routes using a priority score derived from a heuristic function. This score considers both the depot-to-customer distance and the vehicle's current position relative to the customer, weighted by parameters α and

beta. As customers are added to routes, feasibility checks ensure that constraints like vehicle capacity, time windows, and maximum route length (epsilon) are not violated, maintaining the feasibility and efficiency of each route.

Throughout the initialization process, feasibility assurance is achieved through constraint verification, where the algorithm continuously checks for constraint violations. If any violations are detected, repair mechanisms adjust depot selections, vehicle assignments, or customer allocations to restore feasibility. Additionally, diversity maintenance ensures a varied set of initial solutions, enabling the algorithm to explore different regions of the solution space and enhancing its ability to avoid local optima and converge toward globally optimal solutions. Initialization strategies that incorporate heuristic methods, such as proximity-based assignments, are known to enhance the quality of the initial population by embedding domain-specific knowledge (Clarke & Wright, 1964; Eiben & Smith, 2015). Diverse and high-quality initial solutions can significantly reduce the number of generations required for convergence and improve the likelihood of discovering optimal or near-optimal solutions (Laporte et al., 2000; Goldberg, 1989).

4.4 Fitness Evaluation

The fitness evaluation step in the GA assesses the quality of each individual (chromosome) in the population, based on predefined objectives and constraints. In the context of the MDLRPTW, this evaluation encompasses two primary metrics, total operational cost and maximum travel distance.

Total operational cost considers multiple cost components. First, depot operating costs represent the cumulative expense of keeping selected depots open, with each depot incurring a fixed annual cost to cover infrastructure and operational expenses. Vehicle operating costs are calculated by multiplying the number of vehicles in use by a fixed cost per vehicle, accounting for maintenance, driver salaries, and other expenses associ-

ated with the fleet. Lastly, travel costs are based on the total distance traveled by all vehicles, adjusted by energy rates and cost per unit distance (electricity price), to capture travel distance cost. It is important to realize that each of these components represents the overall economic impact of routing and depot configurations offered by each of these solutions.

The second primary metric, maximum travel distance, identifies the longest distance traveled by any single vehicle within a solution. Minimizing this metric is essential for balancing workloads across the fleet, preventing overburdened vehicles, and ensuring service efficiency. By reducing the longest route, the algorithm promotes a more balanced and effective distribution of vehicle assignments. In addition to evaluating these objectives, the fitness evaluation process performs rigorous constraint verification to ensure each solution is feasible. Depot capacity constraints require that the total demand assigned to each open depot does not exceed its capacity, preventing operational inefficiencies and additional costs from overloaded depots. Vehicle capacity constraints enforce that each vehicle's load remains within its designated capacity to avoid overloading, which could lead to higher operational costs or service delays. As a result of time window constraints, vehicles must arrive at each customer within a specific time window, assuring timely service and avoiding late fees. Finally, route length constraints restrict the total distance of each vehicle's route to a maximum limit (ϵ), which prevents excessive operational costs and delays.

To manage solutions that violate any constraints, a penalty mechanism is employed. If a solution (individual) breaches any of the defined constraints, it is assigned a high or infinite fitness value, effectively disqualifying it from the selection pool. This penalty approach ensures that the GA focuses on feasible regions of the solution space, guiding the search process toward optimal and viable solutions. Penalty methods are standard in evolutionary algorithms and help maintain feasibility in optimization problems by steering the algorithm away from unworkable solutions (Coello et al., 2002; Deb et al., 2016).

This technique, widely supported in the literature, enhances the practical applicability of the algorithm, as it balances operational costs and route efficiency while adhering to real-world logistics priorities (Michalewicz & Schoenauer, 1996; Fonseca & Fleming, 1995).

In multi-objective optimization, balancing competing objectives while satisfying constraints is crucial for effective problem-solving. Penalty methods are proven techniques for managing constraints, favoring feasible solutions, and enabling the search process to converge on practical and effective solutions (Deb et al 2016). Furthermore, focusing on operational cost minimization and route efficiency aligns with real-world logistics goals, where reducing expenses and maintaining service quality are essential priorities (Toth & Vigo 2002).

4.5 Genetic Operators

4.5.1 Selection

Selection in the GA is performed using the NSGA-II (Non-dominated Sorting Genetic Algorithm II) method, which is designed to handle multi-objective optimization problems and making it a popular choice for problems like MDLRPTW (Zitzler et al 2000). This selection method is important in maintaining a diverse set of high-quality solutions, making it especially suitable for complex problems like the MDLRPTW (Deb et al 2016). The primary purpose of using NSGA-II is twofold, First, it supports multi-objective optimization by maintaining a diverse set of high-quality Pareto optimal solutions across multiple objectives, such as minimizing operational costs and travel distances. Second, it enhances diversity maintenance by using a crowding distance metric that ensures a well-distributed set of solutions across the Pareto front. This metric prevents the algorithm from converging prematurely to a single solution, allowing a broad exploration of potential solutions (Deb et al 2002).

The NSGA-II process consists of several key steps. First, Non-Dominated Sorting ranks individuals based on pareto dominance. Individuals in the first rank are non-dominated, meaning they are not outperformed in all objectives by any other individuals. The second rank includes individuals dominated only by those in the first rank, and so on. This sorting creates a hierarchy of solutions based on their performance across multiple objectives. Next, crowding distance assignment is used within each rank to measure the density of solutions around each individual in the objective space. This crowding distance helps prioritize individuals in less crowded areas, enhancing solution diversity. Finally, in the selection of the next generation, NSGA-II combines the current population of parents and offspring, sorts them by rank and crowding distance, and selects the top individuals to form the new population. This selection process balances preserving high-quality solutions with maintaining a spread across the solution space. Additionally, NSGA-II's computational efficiency and ability to manage large populations make it well-suited to address the complex, multi-depot structure of MDLRPTW (Laporte et al 2000).

4.5.2 Crossover

Crossover in the GA is implemented using a custom crossover function tailored to the unique chromosome structure of the MDLRPTW problem. The primary goal of this crossover method is twofold: to explore the solution space by combining segments of two parent solutions to create diverse offspring, and to preserve structural integrity by ensuring that the logical connections between depot selections and vehicle assignments remain intact in the resulting chromosomes (Prins, 2004). Custom crossover operators that are specifically designed for a problem's chromosome structure have been shown to enhance the GA's capacity to generate high-quality offspring (Pisinger & Ropke, 2007; Bräysy & Gendreau, 2001). The crossover process consists of three main steps. First, depot selection crossover is performed using a single-point crossover technique on the depot selection vector. A random crossover point is chosen, and the segments beyond

this point are swapped between two parent individuals, resulting in two offspring with mixed depot configurations. Next, vehicle assignment crossover is applied in a similar fashion, where a crossover point is randomly selected on the vehicle assignment list, and the segments beyond this point are swapped between parents. This step ensures that vehicles inherit assignments from both parents, potentially combining beneficial traits from each.

After the crossover, a repair mechanism is employed to address any violations of constraints that may arise. For instance, if an offspring has vehicles assigned to depots that are closed in its depot selection vector, the repair function reassigns these vehicles to open depots and reconstructs routes as necessary, or if a depot might have more customers assigned than its capacity, the repair function identifies customers with the highest demand to reassign them to other depots with available capacity. This repair step is essential for maintaining feasibility in the offspring and ensuring that they can be evaluated effectively in the GA.

In MDLRPTW, where decisions about depot selection and vehicle routing are interdependent, maintaining structural integrity during crossover is crucial for preserving the feasibility of solutions and supporting efficient exploration of the search space (Gendreau et al 2008).

4.5.3 Mutation

Mutation in the GA is performed using a custom mutation function, which introduces random variations to the population, thereby enhancing genetic diversity and promoting solution exploration. This mutation approach is essential for preventing premature convergence and ensuring that the algorithm can explore a broad solution space, potentially discovering optimal or near-optimal solutions that may otherwise be overlooked (Gendreau et al., 2008). Mutation operators play a significant role in maintaining genetic diversity within the population, helping the algorithm avoid local optima by exploring

various parts of the solution space (Gen & Cheng, 1999; Blum & Roli, 2003).

The mutation process involves several steps to modify different parts of the chromosome. First, depot selection mutation is applied to the depot selection vector, where each bit has a specific probability of being flipped. This flipping effectively opens or closes depots, allowing for the exploration of various depot configurations within the solution space. Next, vehicle assignment mutation is performed on the vehicle assignment list. Each vehicle has a chance of being reassigned to a different open depot, which introduces variability in vehicle allocations and encourages the exploration of alternative depot-to-vehicle assignments.

To ensure the feasibility of the mutated individual, a repair mechanism is invoked after mutation. This repair function addresses any constraint violations that may arise, such as vehicles being assigned to depots that are closed after mutation. The repair mechanism reassigns these vehicles to open depots and adjusts routes as necessary, maintaining the structural integrity and feasibility of the mutated solution.

4.6 Repair Mechanisms

Repair mechanisms are important in the GA to maintain a population of feasible solutions throughout the evolutionary process. Given the complexity of the MDLRPTW, which involves multiple constraints, robust methods are needed to address any infeasibilities that may arise during genetic operations like crossover and mutation (Coello, 2002). These mechanisms ensure that the algorithm remains focused on feasible solutions, which is crucial for achieving practical and high-quality solution regions (Gen & Cheng, 1999; Laporte et al., 2000). The process of constraint handling and repair includes several steps to validate and correct solutions post-crossover and mutation. First, the depot validation checks that at least one depot remains open. If crossover or mutation operations result in all depots being closed, the repair function randomly reopens a depot to ensure opera-

tional feasibility. Following this, vehicle reassignment addresses cases where vehicles may have been assigned to closed depots. In such instances, vehicles are reassigned to open depots, ensuring that each vehicle has a valid starting point. Customer reassignment is another critical component of the repair process, which includes two key checks, depot capacity compliance, and vehicle capacity compliance.

In depot capacity compliance, If a depot is overloaded due to customer allocations exceeding its capacity, customers are reallocated to other open depots to restore compliance. In vehicle capacity compliance, customer assignments to vehicles are adjusted to ensure that each vehicle's load remains within its capacity, potentially redistributing customers across vehicles to achieve balanced loads. Following reassignment, route reconstruction ensures the feasibility of routes by rebuilding them to meet constraints, such as vehicle capacity, time windows, and maximum route length. In this stage, heuristic-based reassignment methods are used to assign customers to routes in a way that optimizes route efficiency while maintaining compliance with constraints. This may involve reordering customer sequences or reallocating customers to different vehicles to improve route quality.

The iterative repair process allows the algorithm to handle multiple overlapping constraints. The repair function iterates through adjustments to customer and vehicle assignments until all constraints are satisfied or until a maximum number of repair attempts is reached. Termination of repair is implemented to prevent infinite loops, setting a cap on repair attempts. If feasibility cannot be achieved within this limit, the solution may be discarded or heavily penalized.

4.7 Local Optimization

To enhance the quality of solutions generated by the GA, local search and optimization techniques are applied, focusing on refining individual solutions to improve the overall

effectiveness and efficiency of the algorithm. In the MDLRPTW context, two primary methods ACO and IRE are employed as local search mechanisms. These techniques allow the GA to fine-tune solutions, ensuring high-quality and feasible routes that better meet the problem's objectives.

4.7.1 Ant Colony Optimization (ACO)

ACO is used for route optimization, aiming to refine the sequence of customer visits within each vehicle's route. This optimization seeks to minimize travel distance while adhering to constraints such as vehicle capacity and time windows. Although the problem is not explicitly represented as a graph in our model, the ACO method operates on the underlying structure, which can be conceptualized as a graph $G(V, E)$. Here, V represents locations (nodes, including customers and depots), and E represents connections (edges or routes) between them. The ACO process begins with pheromone initialization (τ_{ij}) on all edges, where pheromone levels are uniformly set across all possible paths. This ensures unbiased exploration by artificial ants during initial iterations. Inspired by the foraging behavior of real ants, ACO algorithms leverage pheromone trails as an indirect communication mechanism to discover optimal routes (Dorigo & Di Caro, 1999). Heuristic information (η_{ij}), typically defined as the inverse of the distance between nodes i and j , guides ants towards closer nodes. It is calculated as:

$$\eta_{ij} = \frac{1}{d_{ij}} \quad (4.6.1.1)$$

Ants make probabilistic decisions based on pheromone intensity and heuristic values, using a probability rule given by:

$$P_{ij} = \frac{(\tau_{ij})^\alpha (\eta_{ij})^\beta}{\sum_{k \in \text{allowed}} (\tau_{ik})^\alpha (\eta_{ik})^\beta} \quad (4.6.1.2)$$

Here, α determines the influence of pheromone levels, while β emphasizes the importance of heuristic desirability. The denominator normalizes the probability across all allowed paths. During route construction, ants prioritize exploration (based on pheromone intensity) and exploitation (based on heuristic desirability), enabling a balance between global search and local optimization (Dorigo & Gambardella, 1997). Once routes are constructed, a Pheromone Update step adjusts the pheromone levels to reinforce successful paths. This involves two components: evaporation and deposition. In the evaporation phase, pheromone levels on all paths decay proportionally, expressed as:

$$\tau_{ij} = (1 - \rho)\tau_{ij} \quad (4.6.1.3)$$

where ρ is the pheromone evaporation rate. This reduction prevents the algorithm from prematurely converging on suboptimal paths. Deposition, on the other hand, strengthens paths taken by ants that discovered shorter or more efficient routes. The amount of pheromone added to these paths is inversely proportional to the route length L , given by:

$$\Delta\tau_{ij} = \frac{Q}{L} \quad (4.6.1.4)$$

$$\tau_{ij} \leftarrow \tau_{ij} + \Delta\tau_{ij} \quad (4.6.1.5)$$

Here, $\Delta\tau_{ij}$ represents pheromone deposition with including Q that represents the pheromone constant, and L is the total route length. This process of route construction and pheromone updating repeats over multiple iterations, gradually converging the pheromone levels toward optimal routes (Reimann et al., 2004; Gambardella et al., 1999). Evaporation facilitates the removal of suboptimal paths, while deposition reinforces promising paths. Integrating ACO with GA combines the strengths of both methods. GA's ex-

ploration capability complements ACO's route refinement process, resulting in improved optimization performance (Blum & Roli, 2003).

4.7.2 Inter-Route Exchange (IRE)

IRE is a local search technique employed to refine solutions by swapping customers between different vehicle routes. IRE are a form of local search that complements the global search capabilities of GA. techniques such as 2-opt, 3-opt, and other neighborhood search methods have long been applied to vehicle routing problems, offering targeted adjustments that refine solutions (Lin & Kernighan 1973). The primary objective of this method is to improve solution quality by balancing workloads and reducing both maximum route distance and overall operational costs. This process provides an additional layer of optimization beyond the standard genetic operations, enabling a more efficient and balanced distribution of routes across the fleet.

The process of IRE begins with the identification of exchange candidates. Pairs of vehicle routes are analyzed to identify potential customer swaps that could enhance key solution metrics, such as minimizing route distances, balancing vehicle loads, and adhering to time window requirements. This initial step sets up potential exchanges that might improve overall efficiency.

Next, feasibility checks are conducted before performing any customer swaps. These checks ensure that the proposed exchange does not violate critical constraints, including vehicle and depot capacities, time windows, and maximum route lengths. This step is important for maintaining the feasibility of the solution after the exchange, ensuring that the optimization process does not produce infeasible routes.

If a feasible and beneficial swap is identified, the algorithm proceeds with performing the exchange, where customers are swapped between routes. The newly created routes are then evaluated to determine if the swap has led to improvements in metrics like total operational cost and maximum route distance. Finally, this process enters an iter-

ation phase, where the algorithm explores multiple exchange possibilities in a repeated manner, continuing until no further beneficial exchanges can be found. This iterative refinement allows the algorithm to escape local optima and enhances the overall quality of the solution (Osman 1993; Lin 1965).

4.8 Evaluation of offspring

An evaluation of offspring is one of the most important steps of GA that help determine the quality of the new solutions that are generated after genetic operations and local optimizations have taken place. Each offspring is assessed based on two key objectives: total cost and maximum distance. The total cost includes the operating costs of open depots, vehicle costs, and the cost of travel distances, calculated using parameters such as depot operating costs, vehicle costs, energy rates, and energy costs per unit distance. These components are aggregated to provide an overall measure of economic efficiency. The maximum distance minimizing the longest route taken by any vehicle, ensuring that no vehicle exceeds the maximum allowable route length. Additionally, in calculating these objectives, the evaluation process checks constraints to ensure that each solution is feasible. The total demand assigned to each depot must not exceed its capacity, and each vehicle must operate within its load capacity. Deliveries to customers are required to occur within specified time windows, and all routes must be constructible based on the distances and travel times between locations. If any of these constraints are violated, the fitness function assigns an infinite fitness value to the offspring. Therefore, by penalizing infeasible solutions in this way, infeasible solutions are prevented from being chosen for further generations.

4.9 Update population

The update population step is a fundamental component of the GA, responsible for forming the next generation of solutions by integrating the newly evaluated offspring into the population. After evaluating the fitness of the offspring, the algorithm updates the population by replacing less fit individuals with the new, more promising solutions. In our implementation, we utilize the NSGA-II, which employs an elitist strategy to preserve high-quality solutions across generations. NSGA-II's crowding distance mechanism ensures a balance between convergence toward the pareto optimal front and the maintenance of diversity within the population (Deb et al., 2002). By integrating evaluated offspring into the population while preserving elite individuals, the algorithm effectively explores the solution space and avoids premature convergence, ultimately leading to optimal or near-optimal solutions.

4.10 Evolutionary Loop

The evolutionary loop is the iterative core of the GA, where the population evolves over multiple generations to improve solution quality for the MDLRPTW. This process includes several key steps like selection, crossover, and mutation enhanced by local search mechanisms that collectively guide the optimization toward feasible and high-quality solutions (Corne et al 1999; Blum & Roli 2003).

The loop begins with selection, where the NSGA-II method is used to choose diverse and high-quality parent individuals based on pareto dominance and crowding distance. Additionally, the use of NSGA-II within the evolutionary loop facilitates effective handling of multi-objective trade-offs between pareto optimal solutions (Deb et al., 2002; Deb et al., 2016). This step ensures that the best solutions are propagated to the next generation while maintaining population diversity. Following selection, crossover and mutation are applied to the selected parents. The custom crossover operation combines genetic

material from both parents, and mutation introduces variability to explore new areas of the solution space.

Once crossover and mutation are completed, a repair step is undertaken to enforce feasibility. This mechanism adjusts depot selections, vehicle assignments, and routes as necessary, maintaining the structural integrity of each solution. The repaired offspring then undergo local search optimizations, where ACO refines vehicle routes, and IRE enhances balance across routes, further improving solution quality.

After the local search, fitness evaluation assesses each new individual based on total operational cost and maximum route distance, with penalties applied to any solutions that fail to meet constraints. In the population replacement phase, the evaluated offspring are integrated into the population, often replacing less fit individuals. NSGA-II's elitist mechanism preserves high-quality solutions. The elitist strategy in GA is a mechanism designed to ensure that the best solutions (individuals) found during the evolutionary process are preserved and carried over to subsequent generations. This strategy prevents the loss of high-quality solutions due to the stochastic nature of genetic operators (Bäck, 1996). Finally, a termination check is performed to determine if the loop should continue. The loop repeats until predefined stopping criteria such as a set number of generations or a lack of improvement are met, allowing the algorithm to stop once satisfactory solutions have been identified.

4.11 Solution Extraction

Solution extraction represent the final stages of the algorithm, where the best solutions are identified, analyzed, and presented in a comprehensible format. This step is crucial for interpreting the results, validating the optimization process, and providing actionable insights to stakeholders.

4.11.1 Best Solution Identification

The Hall of Fame component maintains a record of top-performing individuals throughout the evolutionary process. By storing these elite solutions in the HOF, the algorithm ensures that the best solutions are not lost in subsequent generations, allowing continuous refinement while preserving high-quality outcomes.

Given the bi-objective nature of MDLRPTW, pareto front extraction is utilized to identify multiple pareto-optimal solutions. Each solution on the pareto front represents a unique trade-off between total operational cost and maximum route distance. This approach provides a range of solutions, allowing decision-makers to select an option that best aligns with their specific priorities and operational goals.

The algorithm performs a detailed solution analysis to provide insights into the best solution's structure. This includes listing open depots to show depot utilization, detailing vehicle assignments to illustrate fleet distribution, and outlining customer routes for each vehicle to ensure all demands are met within time windows. Arrival times at each location confirm time window compliance, enhancing service reliability. This analysis helps stakeholders understand solution operations and logistical compliance.

Finally, performance metrics are presented to offer quantitative insights into the effectiveness of the solution. The total operational cost, which includes depot operating costs, vehicle operating costs, and travel costs, provides a holistic view of the solution's economic efficiency. Additionally, the maximum route distance highlights the longest distance traveled by any vehicle, offering a perspective on load balancing and operational efficiency.

Chapter 5

Computational Results

5.1 Introduction

In this chapter, we present the validity of the model and the exact solution for small sizes, then we will go over the HGA for solving in large scale of the model to indicate the pareto set of solutions. In pareto set solution we will find some solutions to check different trade-offs between two objectives. These pareto solutions allow us to start sensitivity analysis on them to see how the model will change in different objectives and outcomes. All computational experiments were conducted on a computer with 16 GB of RAM and an Intel Core i5 CPU (2.5 GHz). NSGA-II is one of the most commonly used algorithms to solve multi-objective optimization problems, and it has been implemented in Python using a library called DEAP (Distributed Evolutionary Algorithms in Python)(Fortin et al 2012).

5.2 Model Validation

To validate the accuracy and effectiveness of the developed model, a comparative analysis was conducted between the CPLEX solver in GAMS and a metaheuristic HGA to get a better understanding of their efficiency and effectiveness. This comparison was

performed on small-sized instances of the MDLRPTW for EVs. The primary objectives were to minimize both the total cost and the maximum travel distance. The model was tested on five different instances, each with varying depot and vehicle assignments. The results are summarized in Table 5.1 Each row shows the total cost and maximum distance for the solutions provided by CPLEX and the GA, along with the percentage gap. It can be seen from Fig. 5.1 that Cplex has better performance when dealing with

	Objective function	C8,D2,K2	C10,D5,K3	C11,D3,K3	C12,D5,K3	C15,D5,K4	C18,D7,K7
Cplex	Total Cost	70.64	87.99	130.1	134.86	133.05	154.69
	Max Distance	92.28	169.6	72.11	96.8	97.76	74.4
Genetic	Total Cost	80.59	88.97	131.89	138.99	135	157.33
	Max Distance	94.03	161.51	70.96	101.15	98.1	78.24
GAP %	Total Cost	1.1%	1.1%	1.37%	3.06%	1.48%	1.7%
	Max Distance	1.8%	5%	1.8%	1.55%	1%	5.1%
Time Elapsed	Cplex	5 sec	15 sec	8 sec	156 sec	812 sec	19977 sec
	Genetic	87 sec	89 sec	95 sec	91 sec	120 sec	249 sec

Table 5.1: Comparison of CPLEX and Genetic Algorithms for Various Instances

smaller models than HGA, but when the problem size increases, it struggles significantly. For smaller configurations like "C8, D2, K2," cplex completes calculations quickly (5 seconds), but its computational time escalates steeply for larger configurations, reaching 19,977 seconds (approximately 5 hours and 33 minutes) for "C18,D7,K7." HGA, in contrast, demonstrates consistent computational times across all configurations, with a maximum of 249 seconds (about 4 minutes) for the largest model, significantly outperforming cplex on larger models. This trend highlights HGA's scalability and suitability for handling complex, large-scale MDLRPTW models, while cplex is more practical in smaller instances where computational demand is lower.

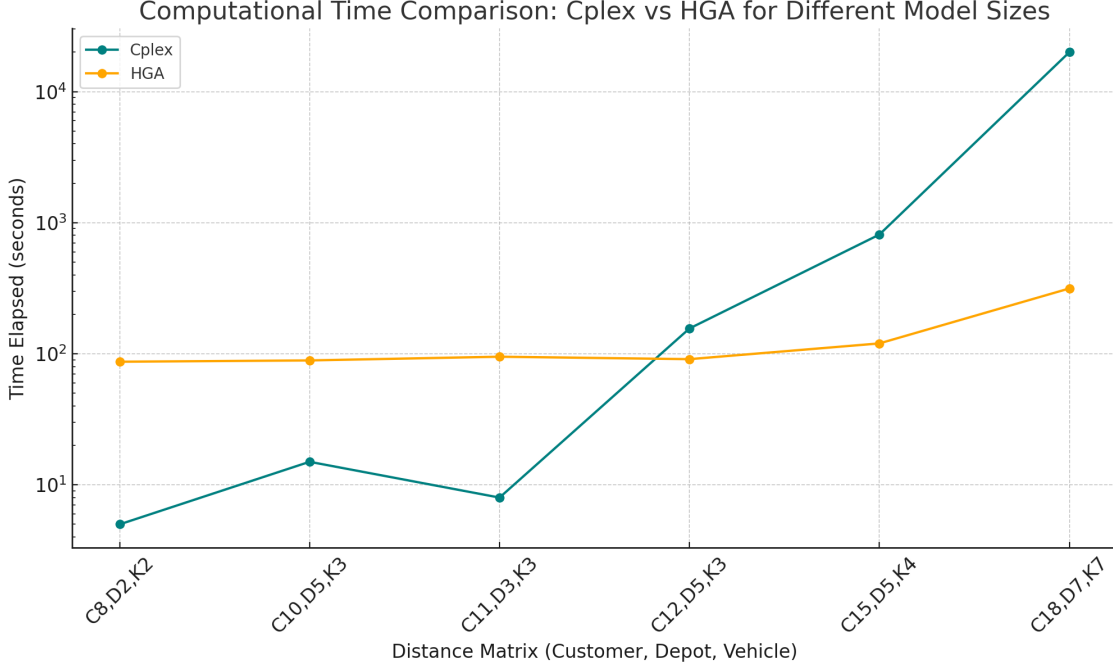


Figure 5.1: Computational Time

In terms of solution quality in Table 5.1, the gap between the total cost and maximum distance for CPLEX and HGA remains generally acceptable, with most differences under 2%. Only three instances show a gap between 3% and 5.1%. Specifically, in the example with 18 customers and 7 depots, HGA achieves a maximum distance with a gap of only 5.1% and a total cost with a gap of just 1.7% compared to cplex. As a result of these small differences, HGA is a reliable and efficient alternative to the exact cplex solution, especially for larger instances where computational time becomes a crucial constraint, such as large data sets. The GAP percentage is calculated based on the results of HGA and cplex as follows:

GAP percentage formula

$$GAP = \frac{HGA - Cplex}{Cplex} \times 100$$

We test the model in different large sizes to see how the model works and how computa-

tional time still increases in HGA by increasing the size of the model. For this approach, based on the table. We will use practical parameters based on literature and real-world data from reliable websites and sources. The results are generated based on Table 5.2, Table 5.3 illustrates the results of different large-size of problems. We run the model of 50 customers, 12 potential depots, and 10 vehicles to analysis the model in different scenarios in the next parts. The model network is shown at Fig 5.2

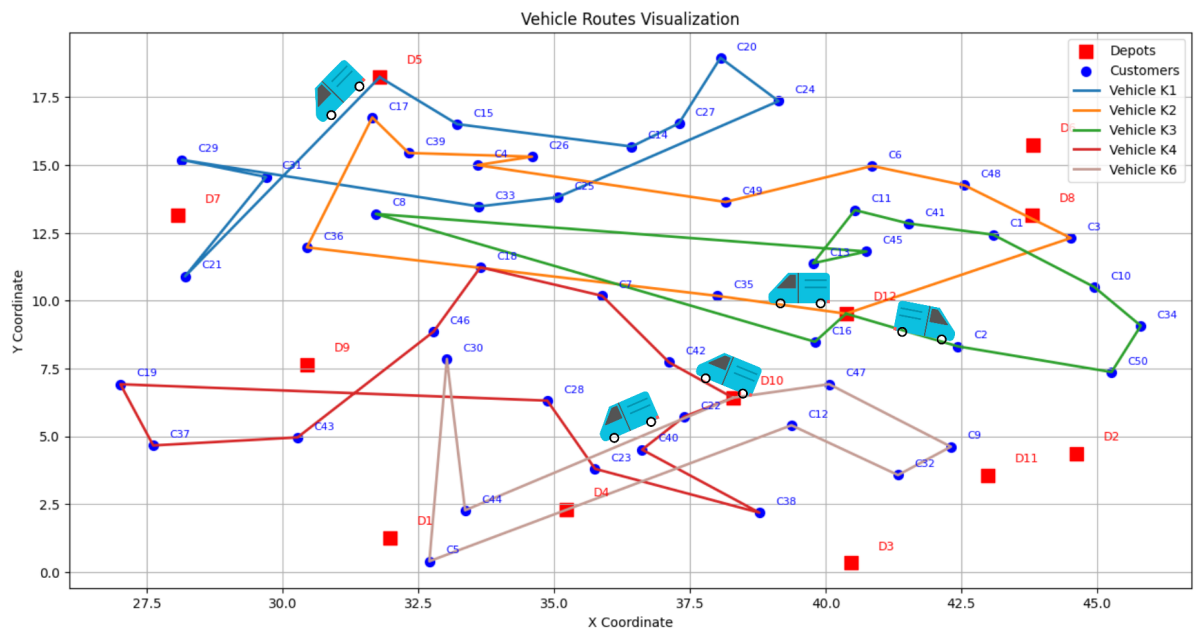


Figure 5.2: Model Network

Notation	Definitions
A_o	Depot capacity.
DE_j	Customer demand.
Q_k	Vehicle capacity of EVs.
Vc	Vehicle fixed cost and labor cost.
yc	Fixed number of working days in year
S_i	Service time for each customer.
$T_{start,i}$	Earliest time window for each customer.
$T_{end,i}$	Latest time window for each customer.
T_{max}	Maximum allowable time for last-time delivery.
D_{ij}	Distance between locations.
T_{ij}	Travel time between locations based on speed.
p	Depot installation cost with charging piles.
elc	Electricity price in Ontario.
e_{rate}	Energy consumption rate for EVs.
ϵ	Vehicle total kilometers.
$Speed$	Average vehicle speed.
n	Optional maximum number of depots selected from potential depot locations.

Table 5.2: Definitions of Parameters

We illustrate how the model works for 50 customers, 12 depots, and 10 vehicles in Fig 5.2. There are 12 potential depots, of which three are opened, D5, D10, and D12. Vehicle K1 starts its route from D5, visiting C21, C31, C29, C33, C25, C24, C20, C27, C14, and C15 in sequence. Vehicle K2 starts its route from D12, visiting C35, C36, C17, C39, C26, C4, C49, C6, C48, and C3. Vehicle K3 also starts from D12, visiting C2, C50, C34, C10, C1, C41, C11, C13, C45, C8, and C16. D10 is opened and assigned to vehicle K4, which begins its route with C42, C7, C18, C46, C43, C37, C19, C28, C23, C38, C40,

HGA Config	C25, K7, D6	C40, K10, D12	C50, K10, D12	C60, K12, D15	C70, K12, D15	C80, K15, D18	C90, K15, D20	C100, K15, D20	C110, K18, D23
Depot, Vehicles used	K3, D1	K4, D2	K5, D3	K7, D4	K8, D5	K9, D4	K8, D4	K11, D5	K13, D7
Total Cost (\$)	256,893	430,550	690,700	871,521	1,062,194	1,040,179	922,318	1,189,302	1,579,146
Max Distance (km)	51.28	39.74	38.77	34.58	33.89	33.63	41.33	33.37	27.81
Time Elapsed (s)	438.62	671.76	706.79	917.06	1,145.03	1,599.41	1,921.06	2,194.66	2,846.50

Table 5.3: Performance Metrics of Heuristic Genetic Algorithm (HGA) for Large-Scale Instances

HGA Parameter	Number
Crossover prob	0.6
Mutation prob	0.9
Population size	20
Generation number	50
Ants	30
Iteration of ACO	100
Evaporation rate	0.6
Pheromone deposit factor (Q)	200

Table 5.4: HGA Parameters

and C22. following that vehicle K6 starts its route, visiting C44, C30, C5, C12, C32, C9, and C47. In summary, three depots have been opened, and four vehicles have been used to deliver to all 50 customers. As shown in table 5.2, the values of each parameter are detailed here. The depot capacity A_o ranges from 1500 to 3500 kg, based on Allen et al. (2012). The demand for each customer DE_j varies between 50 and 200 kg (Cherrett et al., 2012). The vehicle capacity Q_k is set at 1400 kg, based on the specifications of the Ford E-Transit Van (Ford Motor Company, 2024). The vehicle usage cost Vc is \$40,000, reflecting the average driver's salary (Talent.com, 2024). Following, that to find the total travel cost of each vehicle in a year we should multiply by yc which is 365. The service time for each customer S_i is assumed to be 10 minutes, as suggested by Marius Solomon (1987).

The starting time at the depot is 9:00 AM, which is considered as zero $T_{start,i}$. The earli-

est time a customer can be served ranges from 0 to 60 minutes (9:00 to 10:00 AM), and the latest service time $T_{end,i}$ spans from 300 to 480 minutes (2:00 to 5:00 PM), ensuring that all deliveries are completed by 5:00 PM T_{max} (Kallehauge et al., 2005). The distance between each location (D_{ij}) is set at 20 km (Pourvaizir et al., 2024), as this range can cover most parts of the city of Toronto. For example, assuming a central location in midtown, it would encompass both downtown and uptown areas. while the travel time T_{ij} is calculated based on this distance and an average speed of 35 km/h.

The depot installation cost p , including charging piles, ranges from \$120,000 to \$180,000 (Natural Resources Canada, 2021). The electricity price elc follows the ultra-low overnight rate applicable in Ontario (11:00 PM to 7:00 AM) at 2.8 cents per kWh (Hydro One, 2024). The vehicle electricity consumption rate e_{rate} for the Ford E-Transit van is 0.35 kWh per km (Ford Motor Company, 2024). The maximum distance each vehicle can travel is constrained by ϵ is 256 km(Ford Motor Company, 2024). Finally, n represents the optional maximum number of depots to be opened, which can vary based on the size of the problem. Table 5.3 demonstrates that, as the problem size increases such as 110 customers and 23 depot potential location, the HGA maintains an efficient computational time. We determined that a population size of 20 and a generation count of 50 provided a good balance between solution quality and computational efficiency. I tested each instance 10 times to verify the consistency of the results. This approach enhances the reliability of the findings.

In the HGA, all parameters and costs remain the same as the base model, with the addition of specific parameters for HGA and ACO, such as crossover probability, mutation probability, and population size, As detailed in Table 5.4, these parameters were selected to ensure optimal performance after testing with varying population sizes and generation numbers while considering computational time. Increasing the population size and the number of generations can increase computational time without necessarily improving the quality of the solution beyond a certain point. The crossover and mutation

probabilities were fine-tuned to enhance exploration and prevent premature convergence. Furthermore, ACO parameters were calibrated to improve routing decisions. Full details are presented in Table 5.4.

5.3 Analyzing the Pareto optimal solutions

Table 5.5 displays the Pareto front solutions generated by NSGA-II to understand trade offs between minimizing the total cost and minimizing the maximum distance. The pareto front consists of 9 solutions, with a range of 31.48 km to 58.54 km, and costs ranging from \$578,242 to \$1,006,301. These solutions represent a spectrum of trade-offs between maximum distance and total cost and can valuable insight for decision-makers in this context. In order to facilitate further analysis, we have identified three distinct solutions within the pareto front based on their objective function values in order to facilitate this further analysis. Solution S1 and S9 with max distance of 58 km and \$578,242 total cost, max distance of 31.48 km and \$1,006,301 total cost respectively. Comparing S1 and S9 indicates that by investing \$428,059 we can reduce the maximum distance significantly. However, it seems that there is a need to add two more depots and to use more vehicles in order to decrease the maximum distance at a time when this seems to be a capital-intensive undertaking. Another significant observation emerges from the midpoint of Fig.5.3, S4 where a moderate investment leads to a substantial reduction in maximum distance and compared to S9 decreases \$321,809 which is a considerable deduction. As a result, this emphasizes the importance of assessing both the balance between total cost and maximum distance during the decision-making process, because even minor adjustments to the investment can have a considerable impact on the performance of the system.

These points are considered optimal trade-offs between minimizing the total cost and minimizing the maximum distance. In Table 5.5, we can see that if we want to decrease

Pareto Solutions	Total Cost(\$)	Maximum Distance km
S1	\$578,242	58.51
S2	\$588,032	57.04
S3	\$634,340	44.13
S4	\$656,740	39.92
S5	\$690,700	38.77
S6	\$700,753	38.04
S7	\$733,261	35.76
S8	\$841,613	33.43
S9	\$1,006,301	31.48

Table 5.5: Pareto Set

the maximum distance, we should use more depots or vehicles, which increases the cost. However, if we want to reduce costs, we should install fewer depots and minimize the number of EVs, which increases the maximum distance. Trade-offs between both objectives can be observed at three points: S1, S4, and S9. Point S1 is considered due to high cost and low maximum distance, while S9 reflects low cost and high maximum distance. S4 represents a balance between these two, meaning both objectives are minimized and accounted for, resulting in a solution that is not excessively expensive and avoids significant unfairness between drivers.

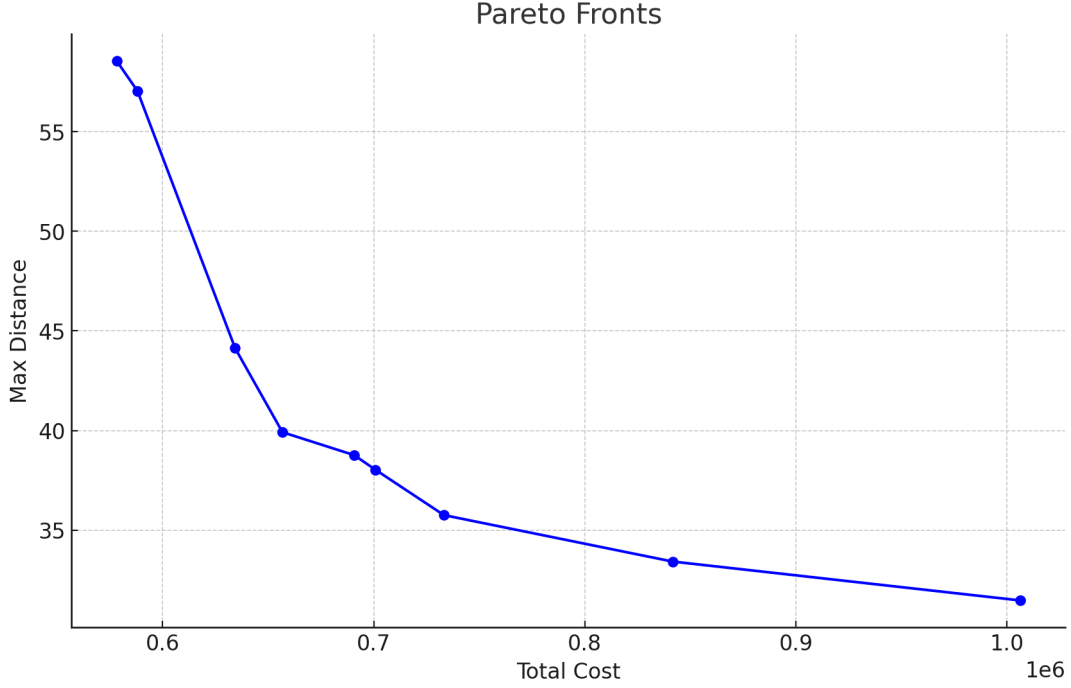


Figure 5.3: Pareto Fronts of HGA.

5.4 Sensitivity Analysis

As part of our study, we conducted a detailed sensitivity analysis of the pareto front's performance, with a specific focus on three key solutions: S1, S4, and S9. These solutions were carefully selected using the sampling from the pareto edges and center method, a mixed selection technique that combines deterministic and tournament-style approaches. This method ensures a balanced representation of critical points on the pareto front, covering both extreme solutions and less pronounced trade-offs.

The selected solutions represent critical regions on the pareto front. Solutions like S1 and S9 correspond to extreme points, where one objective is optimized to the fullest, often at the expense of significant deterioration in other objectives. These points lie at the edges of the pareto front in the objective space, showcasing the boundaries of possible trade-offs.

On the other hand, S4 is a knee point, representing a significant bend or corner in the

pareto front curve (Branke et al., 2004). Knee points are characterized by a dramatic trade-off dynamic: any small improvement in one objective results in a substantial deterioration in at least one other objective. These points are particularly important as they often represent well-balanced solutions that stakeholders might prefer for their compromise between competing objectives. Our goal is to gain a nuanced understanding of how different solutions balance competing objectives under varying conditions. Additionally, this approach highlights the trade-offs between objectives as well as the robustness and practicality of the solutions.

The choice of these specific solutions was strategic. Solution S1 represents a scenario with low costs but a high maximum vehicle distance, useful when minimizing cost is a priority over maximum distance. Solution S4, on the other hand, embodies a balanced approach, offering insights into performance when neither cost nor distance dominates. Lastly, S9 is designed for situations where the primary focus is on reducing the maximum distance, with less concern for cost, and is suitable for applications where distance is the critical factor.

The sensitivity analysis aimed to achieve two objectives: firstly, to assess the stability of these solutions by varying parameters like epsilon (ϵ) and vehicle capacity (Q_k), thus evaluating how resilient each solution is to changes. Secondly, to inform decision-making by understanding which solutions might better handle real-world operational fluctuations. This approach, as discussed in Avila et al. (2006), not only highlights the variability in solution performance but also aids stakeholders in making informed decisions about system configurations in dynamic environments.

5.4.1 The effect of epsilon (ϵ) on objective functions

In this section, a sensitivity analysis was conducted to examine the impact of epsilon on the maximum distance and total cost of the solution. Several scenarios based on three solutions on different epsilons were investigated to observe changes in these parameters.

Specifically, 7 scenarios for epsilon ranging from less than 20 to up to but not including 50 were explored in depth to understand how these increments affect the performance and trade-offs of the solutions.

In Table 5.6, the results show that increasing the epsilon from 20 km to 50 km significantly impacts both objectives. In S1, we evaluated the effect of setting epsilon below 20 km and observed that the total cost increased substantially by 104%, reaching \$1,180,066. For solution S9 we can see an increase of 32.94% reached \$1,337,802 which is because the model tries to make sure the maximum distance for each vehicle did not exceed 20 km. S4 is increased by 51.95% and reached \$997,182. In contrast, Figure 5.4 shows that by relaxing the maximum distance constraint and allowing the model to explore more efficient routes, there is a substantial reduction in total cost. It means that the model will use fewer vehicles or not open new depot near customers to reduce the total maximum distance. It should also be noted that Table 5.7 illustrates that with an increase in epsilon from 20km to 50 km, there is a corresponding increase in the maximum distance as well. This suggests that the HGA prioritizes keeping the maximum distance below the epsilon value, effectively using the maximum distance as a key constraint in its optimization process.

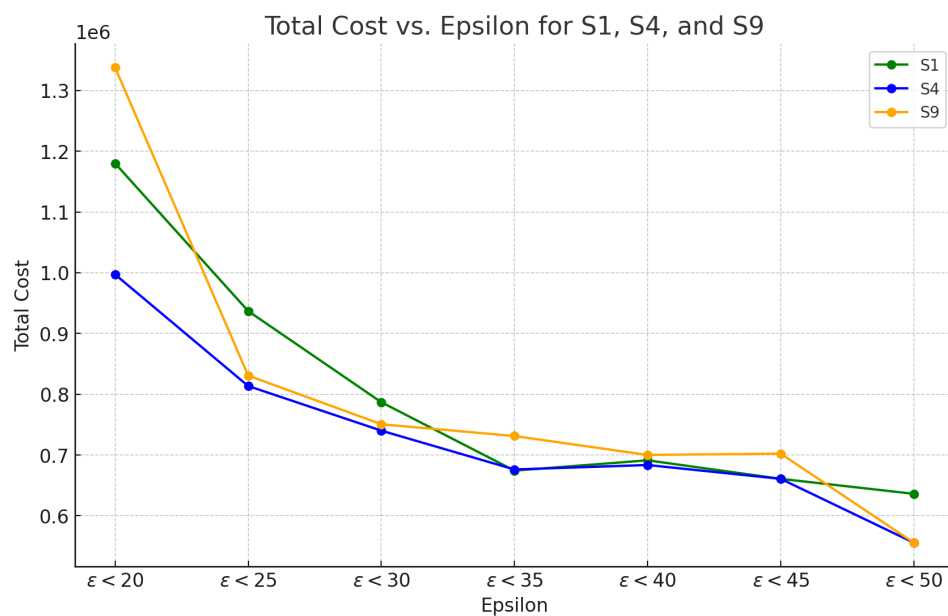
TC of Solutions	$\epsilon \leq 20$	$\epsilon \leq 25$	$\epsilon \leq 30$	$\epsilon \leq 35$	$\epsilon \leq 40$	$\epsilon \leq 45$	$\epsilon \leq 50$
S1	\$1,180,066	\$936,881	\$786,877	\$674,320	\$691,237	\$660,774	\$636,115
S4	\$997,182	\$813,508	\$740,091	\$676,177	\$683,521	\$660,761	\$555,596
S9	\$1,337,802	\$830,742	\$750,651	\$731,231	\$700,091	\$702,227	\$555,528

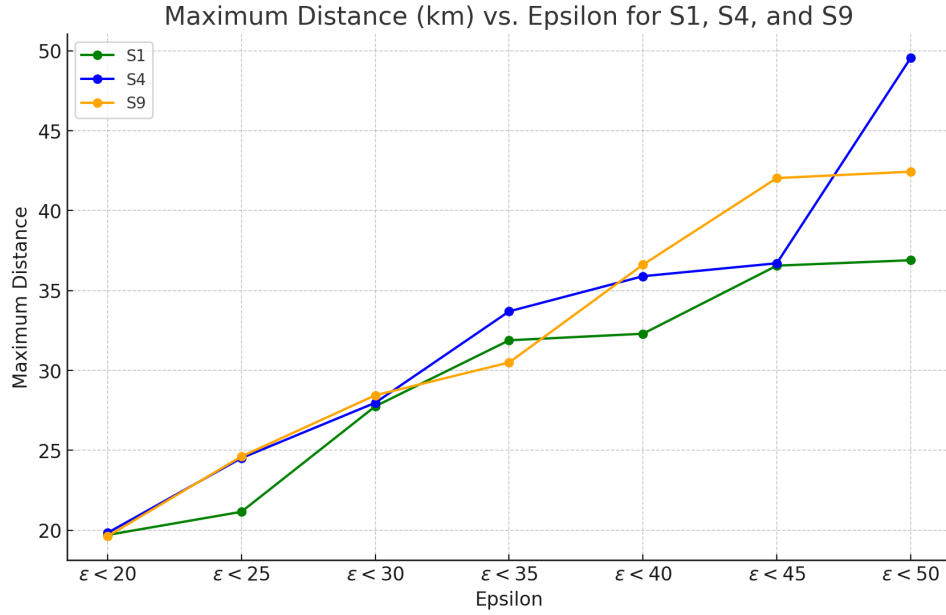
Table 5.6: The effect of epsilon on total cost

MD of Solutions	$\epsilon \leq 20$	$\epsilon \leq 25$	$\epsilon \leq 30$	$\epsilon \leq 35$	$\epsilon \leq 40$	$\epsilon \leq 45$	$\epsilon \leq 50$
S1	19.71	21.16	27.76	31.89	32.30	36.56	36.9
S4	19.84	24.52	27.98	33.70	35.90	36.71	49.56
S9	19.84	24.62	28.45	30.50	36.62	42.04	42.44

Table 5.7: The effect of epsilon on maximum distance (km)

It is evident from Fig. 5.5 that we demonstrate S1, S4, and S9 together to understand that epsilon ϵ has a significant impact on the performance of a system, particularly when there is a need for a shorter maximum distance. By analyzing the data, we can see model wants to use fewer vehicles when we relax Epsilon (ϵ) and it means the maximum distance will increase.

Figure 5.4: The effect of epsilon(ϵ) on Total Cost.

Figure 5.5: The effect of epsilon(ϵ) on Maximum Distance.

5.4.2 The effect of vehicle capacity (Q_k) on objective functions

This section presents a sensitivity analysis assessing the impact of vehicle capacity on maximum distance and total cost. For solutions S1, S4, and S9, we evaluate changes in vehicle capacity to observe its effects on maximum distance and total cost. First, we decreased vehicle capacity (Q_k) 10% then increased 10% to 40% to see changes in maximum distance and total cost. As Table 5.8 shows, by increasing vehicle capacity in each solution cost will decrease.

TC of Solutions	-10% Q_k	0% Q_k	+10% Q_k	+20% Q_k	+30% Q_k	+40% Q_k
S1	\$908,740	\$578,242	\$633,445	\$640,360	\$630,050	\$587,005
S4	\$717,010	\$684,492	\$636,958	\$475,435	\$475,447	\$475,435
S9	\$1,000,909	\$1,006,301	\$660,762	\$632,172	\$643,407	\$611,755

Table 5.8: The effect of vehicle capacity (Q_k) on total cost

MD of Solutions	-10% Q_k	0% Q_k	+10% Q_k	+20% Q_k	+30% Q_k	+40% Q_k
S1	30.24	58.54	54.2	57	58.1	77.1
S4	39.18	39.94	55.65	43.33	51.15	68.70
S9	30.10	31.48	43.18	58.28	57.5	73.86

Table 5.9: The effect of vehicle capacity (Q_k) on Maximum Distance(km)

An analysis of Table 5.8, reveals strategies for reducing costs by exploring the relationship between vehicle capacity and total cost across solutions S1, S4 and S9. Fig 5.6, shows that in each scenario, increasing vehicle capacity generally leads to decreased total costs, though the pattern varies. For example, S4 shows a significant drop in costs up to a 20% capacity increase, after which costs stabilize, suggesting that moderate capacity gains optimize total costs. Similarly, in S1, costs sharply decline as capacity rises but show diminishing returns beyond 30%, implying that further increases offer limited additional savings.

In contrast, S9 presents a more complex pattern, with costs fluctuating as capacity increases. This suggests that while greater capacity can reduce costs, the optimal level depends on specific scenario constraints. Conversely, decreasing capacity results in higher total costs across all scenarios, as vehicles must make more frequent trips to customers or utilize additional vehicles to meet demand. This insight helps identify an optimal capacity level that balances cost savings with operational efficiency, guiding strategic decisions for fleet composition tailored to varying demand environments. An analysis of Table 5.9 of the relationship between vehicle capacity increase and maximum distance across solutions S1, S4, and S9 provides valuable insight into fleet performance and operational strategy. Across all solutions, as vehicle capacity increases, the maximum distance each vehicle can cover generally rises, indicating that greater capacity allows for longer routes without returning to the depot. However, In Fig 5.7 each solution displays unique patterns: S1 shows a more gradual increase, while S4 and S9 exhibit sharp jumps at specific

capacity levels. For instance, S4 and S9 experience significant increases in maximum distance around 30% and 40% capacity increments, which could suggest a threshold at which vehicles become capable of handling longer routes. Conversely, decreasing vehicle capacity reduces the maximum distance each vehicle can cover. Understanding these thresholds is essential for cost-benefit analysis, as it allows fleet managers to assess whether the benefits of increased range outweigh the costs associated with larger vehicles.

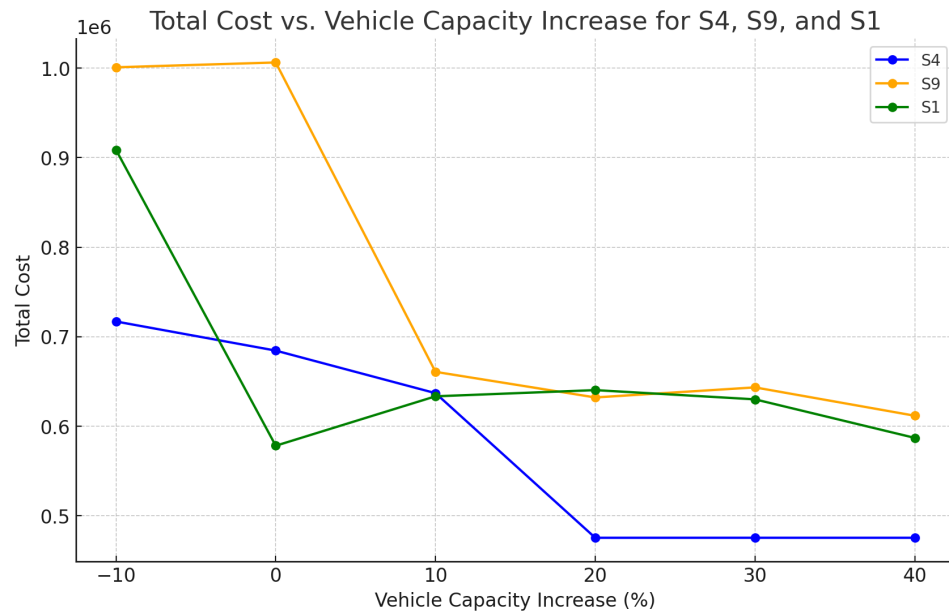


Figure 5.6: The effect of vehicle capacity(Q_k) on Total Cost.

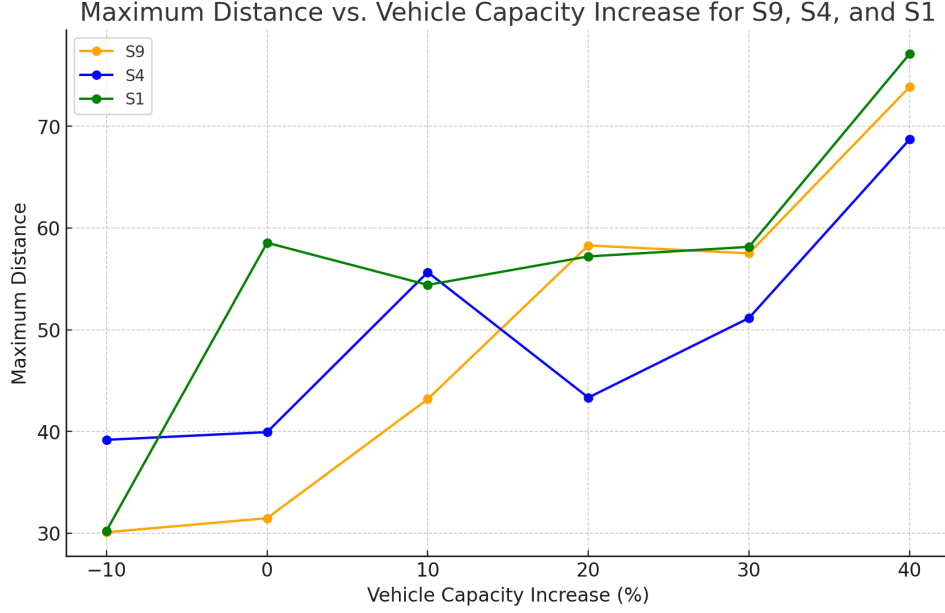


Figure 5.7: The effect of vehicle capacity(Q_k) on maximum distance.

5.4.3 The effect of EVs consumption rate (e_{rate}) on objective functions

This section presents a sensitivity analysis of EV consumption rates to evaluate their impact on total cost and maximum distance. For solutions, S1, S4, and S9, EV consumption rates are varied from a 10% to 50% increase to assess their effect on both objective functions. In Table 5.10, fluctuations in total cost are observed across all three solutions; however, no significant increases or decreases suggest that the model is not highly sensitive to changes in EV consumption rates. In solution S4, the highest single percentage change occurs between 30% and 40%, resulting in a +5.47% increase, while the lowest is a -5.38% decrease between 10% and 20%. For solution S9, the highest fluctuation is +9.75% from 10% to 20%, with the lowest at -4.81% between 20% and 30%. Solution S1 reflects a similar trend, with the highest fluctuation recorded at +4.10% and the lowest at -3.92%. These fluctuations indicate that while changes in EV consumption rates can lead to slight variations in total costs, the overall effects remain minimal. This can be

attributed to the dominance of strategic costs—such as depot installation and vehicle acquisition—over operational costs like travel expenses. The additional cost from EV consumption rate increases is relatively small compared to fixed infrastructure expenses, resulting in low sensitivity to changes in EV consumption rates. Depot and vehicle usage patterns further reinforce this observation. In Solution S1, two depots are opened and six vehicles are used at each increment. For solution S4, three depots remain open across all increments, with five vehicles consistently deployed. In Solution S9, at 10% and 50% increments, five depots are opened and six vehicles are used, while at 20%, 30%, and 40% increments, five depots and five vehicles are utilized.

In Table 5.11, for maximum distance, the sensitivity analysis of solution S9 reveals a 4.43% decrease as EV consumption rates increase from 0% to 50%. The largest fluctuation is a +3.14% increase between 0% and 10%, with the most significant drop of -3.94% occurring between 30% and 40%. The model maintains stable performance by adjusting routes efficiently, reinforcing the observation that consumption rate changes have minimal influence on maximum distance, as depot and vehicle costs drive the overall results. Similarly, solution S4 experiences a 7.36% decrease in total distance. The most notable fluctuation is a +9.49% increase between 0% and 10%, followed by a -14.18% drop between 10% and 20%. Despite these fluctuations, the model remains stable, indicating that EV consumption rates have a limited impact on total cost, with strategic costs continuing to dominate. Solution S1 shows a more pronounced 23.11% decrease in total distance as consumption rates increase. The highest gain of +5.04% occurs between 10% and 20%, while the most substantial drop of -23.71% is observed between 0% and 10%. Although S1 shows greater sensitivity at lower consumption rates, fluctuations stabilize as rates rise. Nevertheless, strategic costs mitigate the overall effect, ensuring consistent performance across different consumption levels.

TC of Solutions	+10%	+20%	+30%	+40%	+50%
S1	\$555,765	\$555,613	\$555,876	\$578,645	\$555,985
S4	\$696,893	\$659,384	\$653,719	\$689,478	\$680,847
S9	\$1,001,065	\$1,098,709	\$1,066,002	\$1,014,753	\$1,017,931

Table 5.10: The effect of EVs Consumption rate on Total Cost

MD of Solutions	+10%	+20%	+30%	+40%	+50%
S1	44.66	46.91	48.62	48.1	45.01
S4	43.73	37.53	40.34	36.38	37.04
S9	34.48	34.42	33.3	33.26	31.95

Table 5.11: The effect of EVs Consumption rate on Maximum Distance (km)

Chapter 6

Conclusions

In this study, we propose an innovative approach using a hybrid genetic algorithm to tackle the MDLRPTW, which is complex and NP-Hard problem often encountered in logistics and supply chain management. This approach integrates depot location and routing optimization, simultaneously assigning customers to EVs at the depot. The allocation of customers to vehicles is based on the capacity of each vehicle, depot, and the specific demands of each customer and time window. The primary objectives are to minimize the total operational costs and reduce the maximum distance traveled by each vehicle, achieving both cost efficiency and equitable workload distribution.

The proposed solution framework consists of two parts, the first part employs the MILP model, solved with the cplex solver, to obtain exact solutions for small problem instances, ensuring optimality and precision. The second part due to the large computational time of the first part, utilizes a metaheuristic approach, specifically a HGA enhanced with ACO, to effectively find near-optimal solutions for larger instances that are computationally challenging. This combination allows the approach to leverage the accuracy of exact methods on smaller scales while harnessing the efficiency of metaheuristic techniques to address larger, real-world problems with greater complexity. Computational experiments confirmed a gap between the exact solution and near-optimal solution the

less than 5.1% for maximum distance and less than 1.7% gap for the total cost.

The analysis highlights that vehicle capacity and the epsilon value play a significant role in influencing total cost and maximum distance. As either vehicle capacity or epsilon increases, the total cost decreases, but the maximum distance traveled tends to rise. Computational experiments show that a 40% increase in vehicle capacity leads to a substantial reduction in total cost, though it nearly doubles the maximum distance. In order to maintain cost efficiency, it is important to ensure fair workload distribution for drivers. In contrast, increasing EV consumption rates has minimal impact on total cost and maximum distance. The model remains stable across different scenarios, as depot installation and vehicle procurement costs dominate operational expenses. This trend holds for most solutions, with the exception of Solution S1, where a 23.71% reduction in total distance is observed with a 50% increase in EV consumption rate. However, even in this case, the impact is not significant enough to suggest that the model is highly sensitive to consumption rate changes.

Overall, the findings indicate that while vehicle capacity and epsilon adjustments drive substantial shifts in cost and distance, EV consumption rate fluctuations have limited influence. This reinforces the robustness of the model, ensuring that cost efficiency and route optimization can be maintained even as EV performance varies.

Future research should examine how to integrate real-time traffic data instead of static travel times. This could enable dynamic routing adjustments. By using real-time and predictive traffic information, the model could adapt routes to avoid congestion, better manage time windows, and optimize energy consumption for EVs in urban areas. Incorporating such data would enhance delivery reliability, particularly in high-density, unpredictable traffic zones. Another research direction could be to consider the MDLRP with time windows and minimize environmental considerations like carbon emissions, as well as explore the social dimension of workload fairness by incorporating driver preferences or fatigue-related constraints into workforce planning.

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